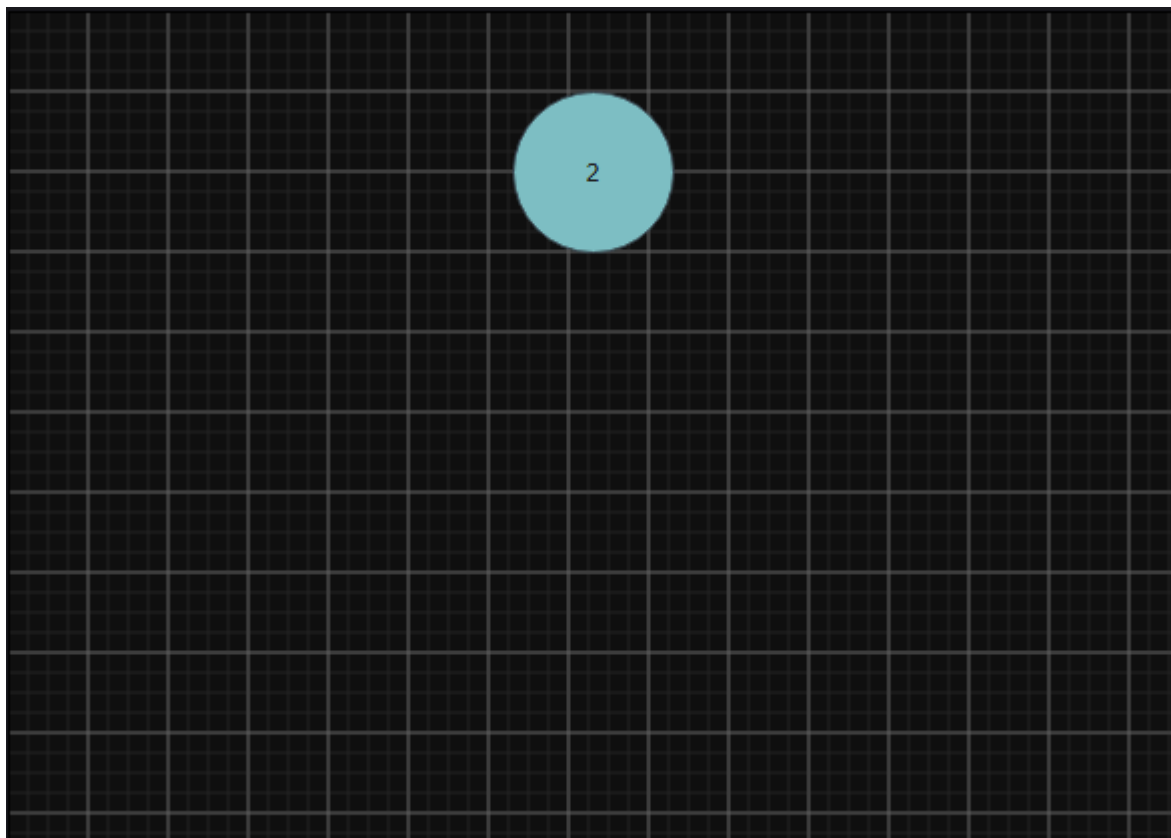
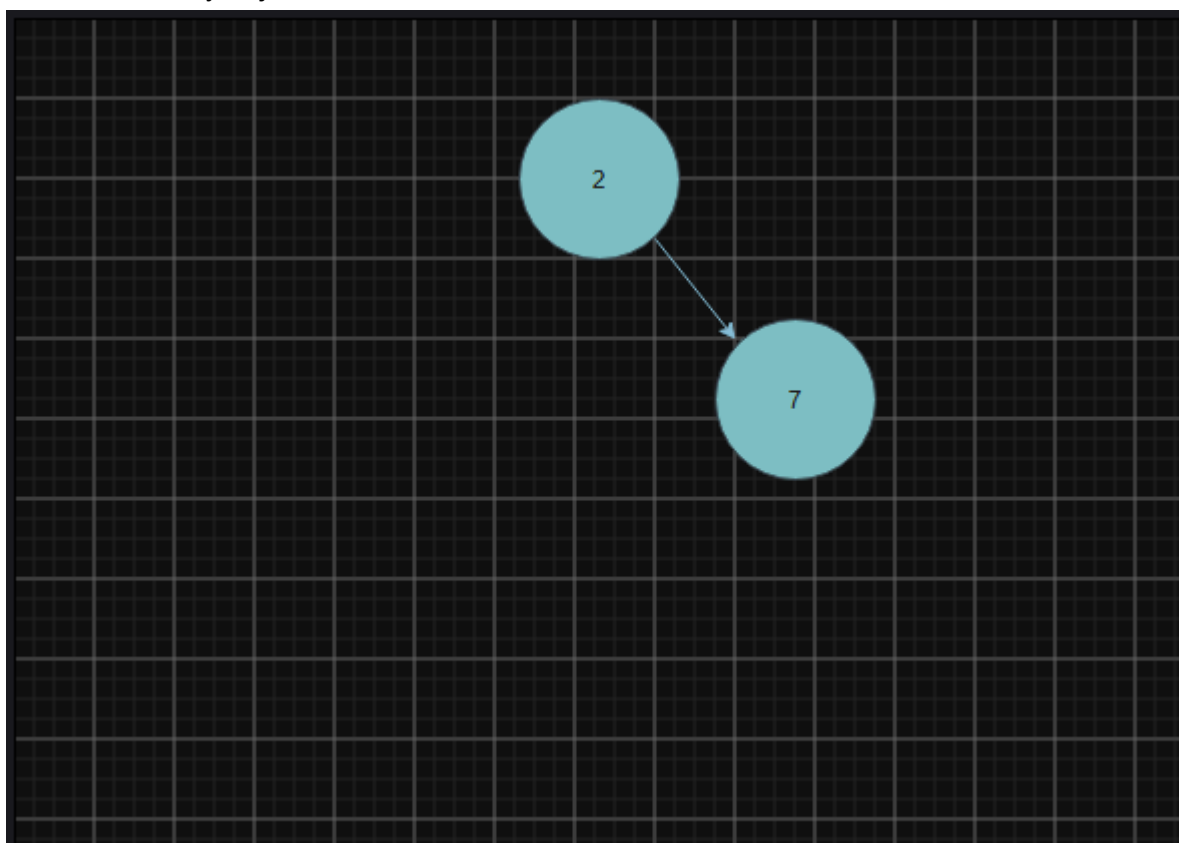


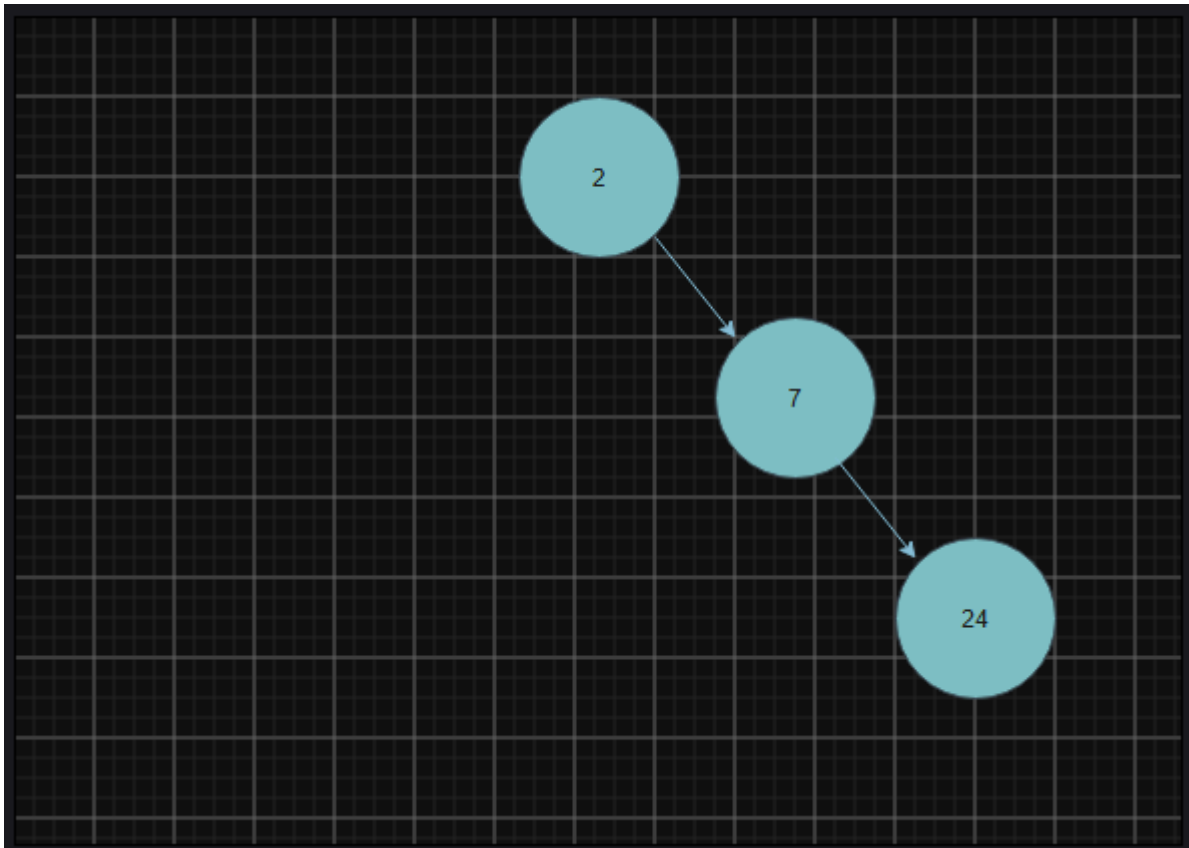
3.a.



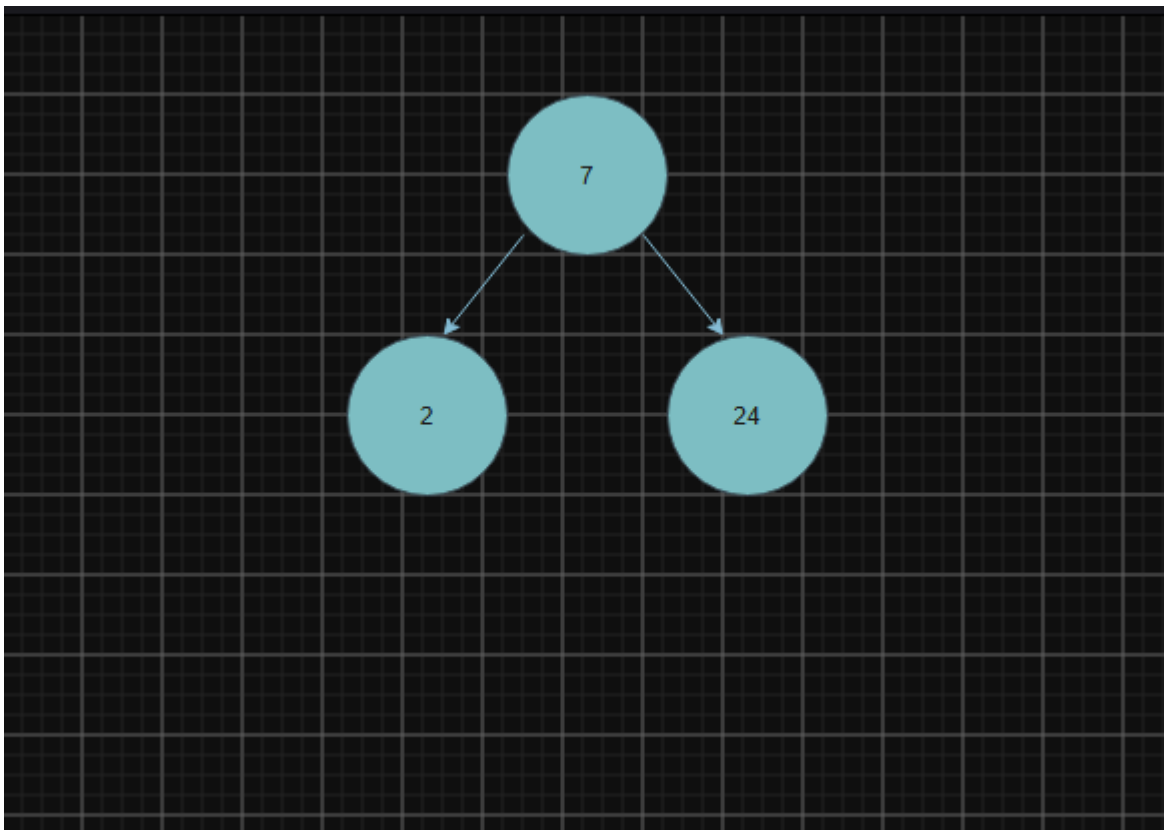
Diff 0. 2 es raíz y hoja.



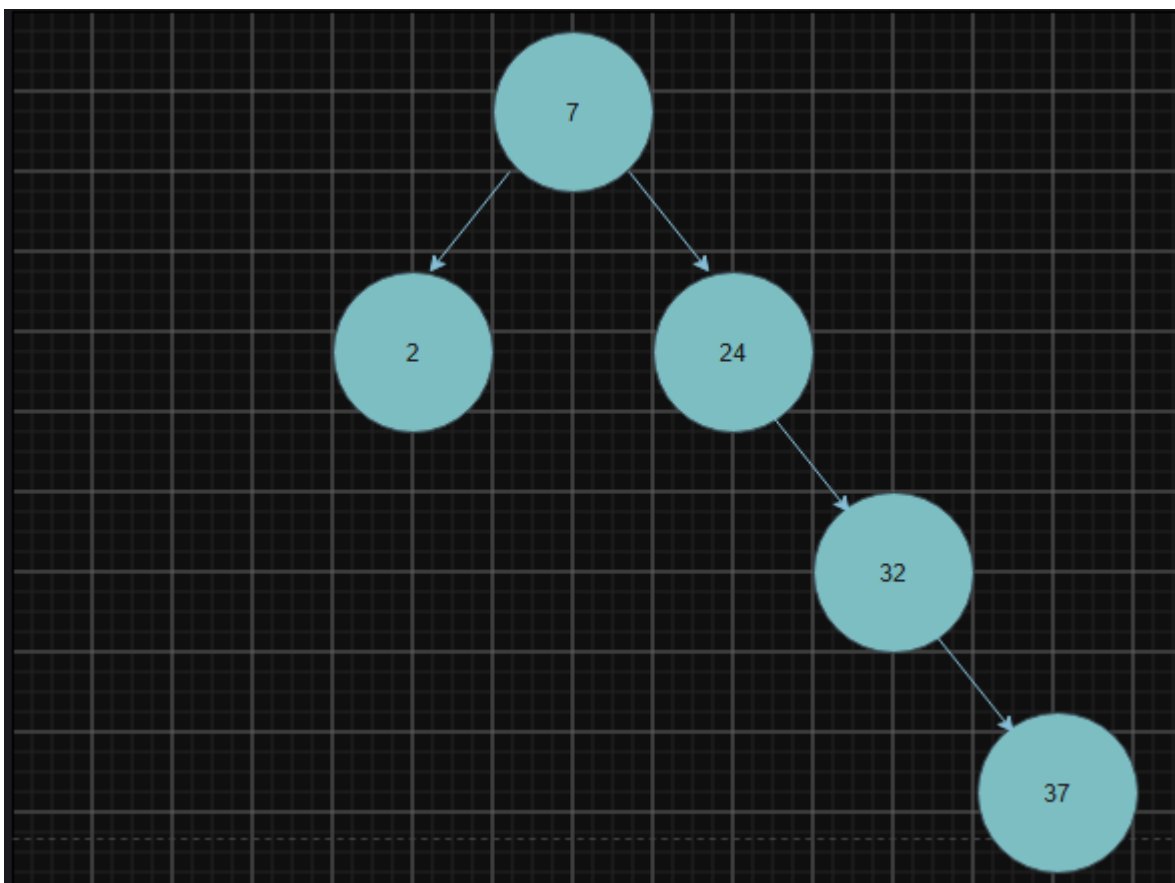
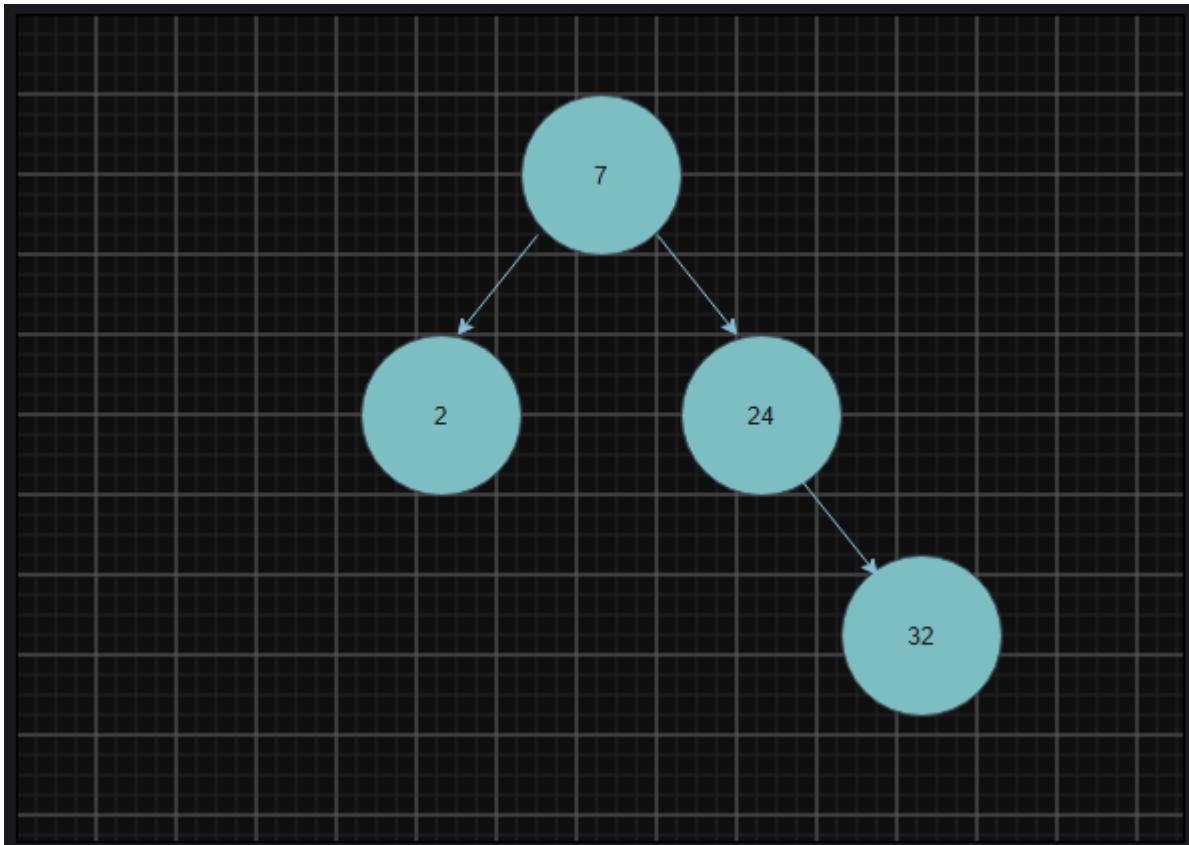
Diff 1. Aun no es necesaria la rotacion



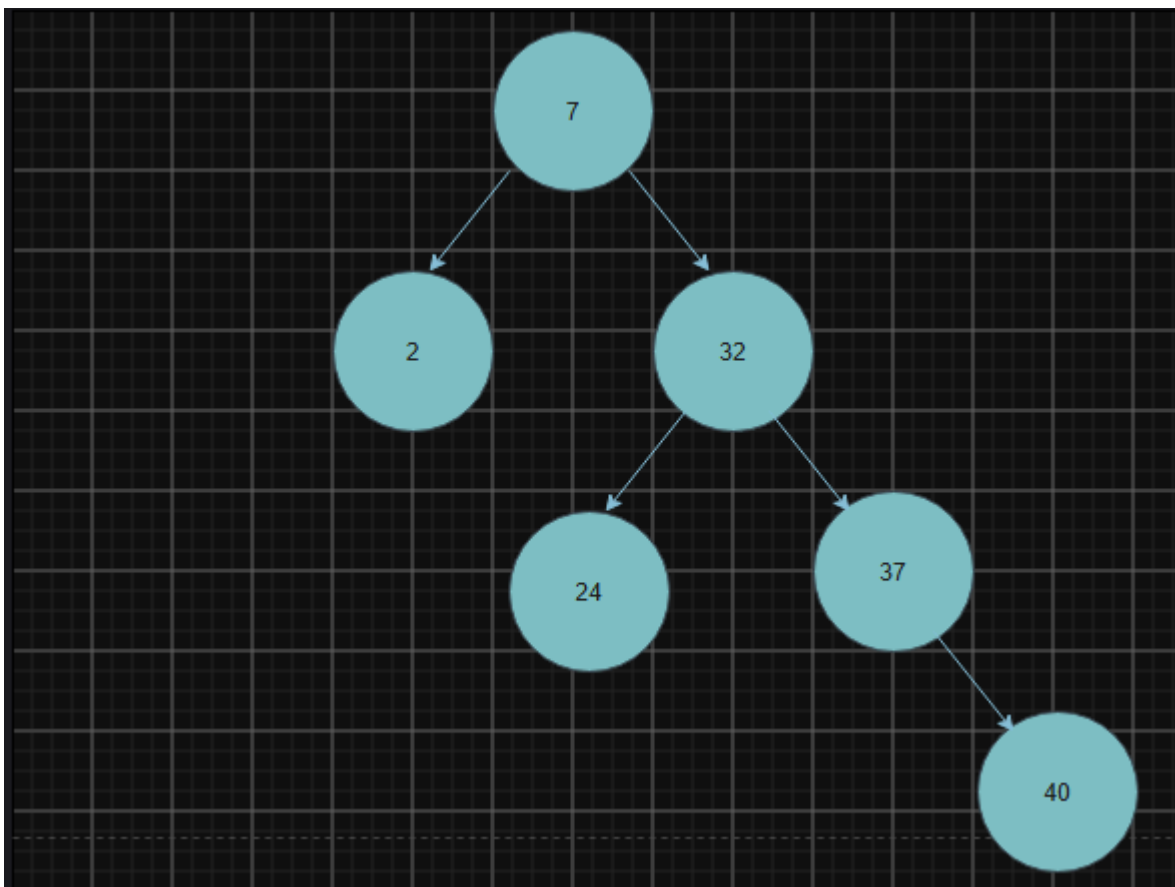
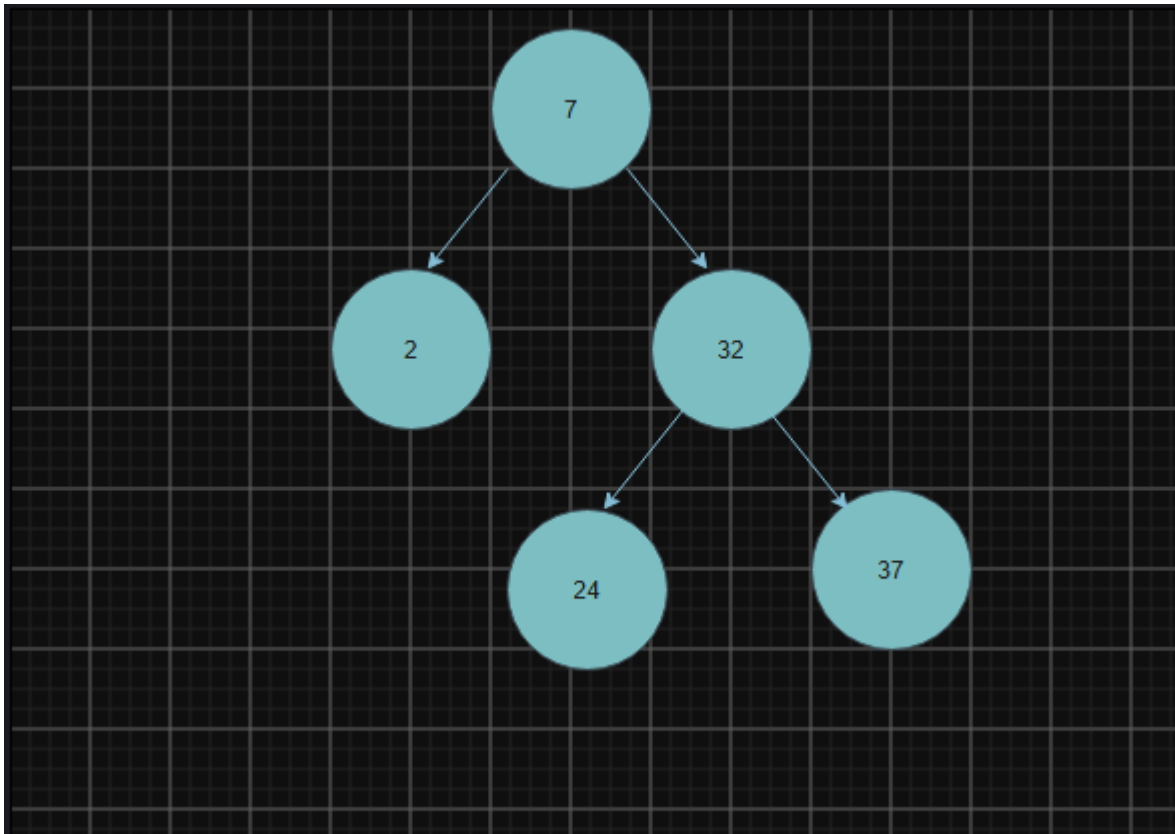
Diff 2. Esta desbalanceado. Rotación Simple Derecha



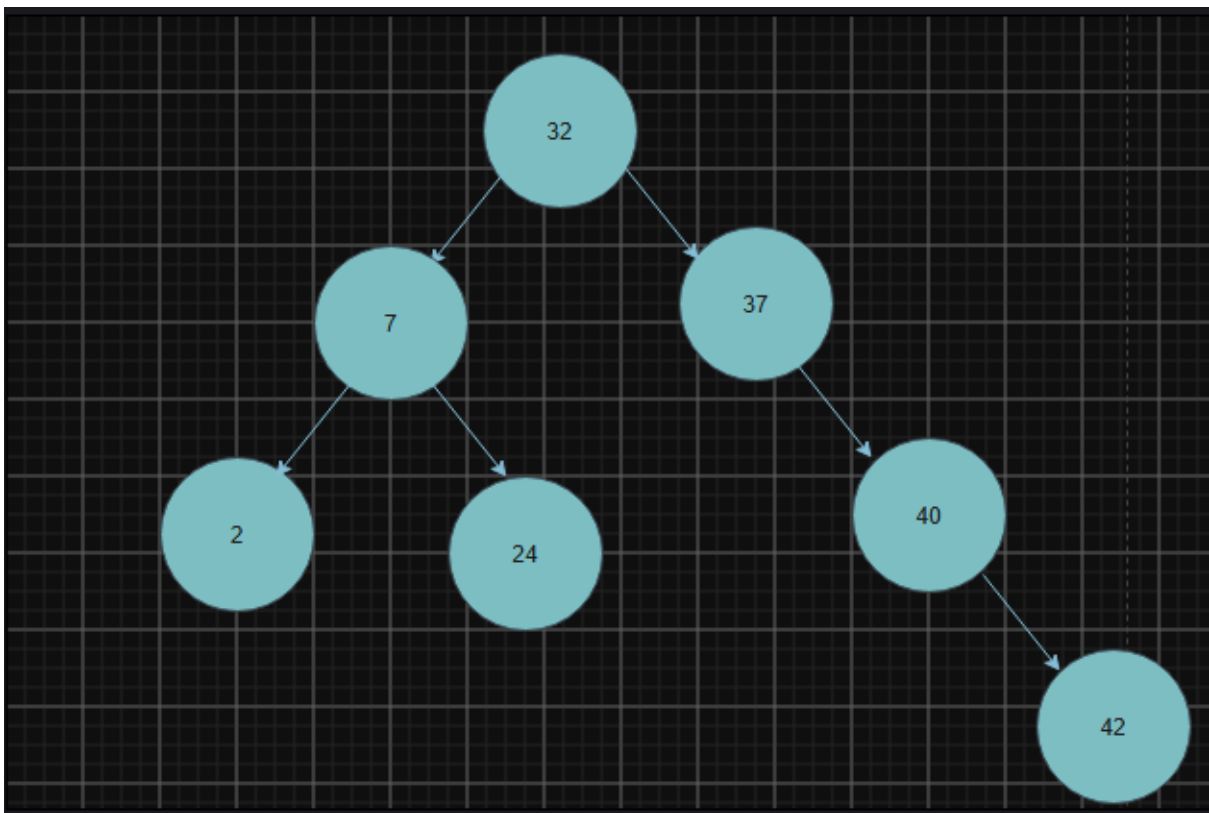
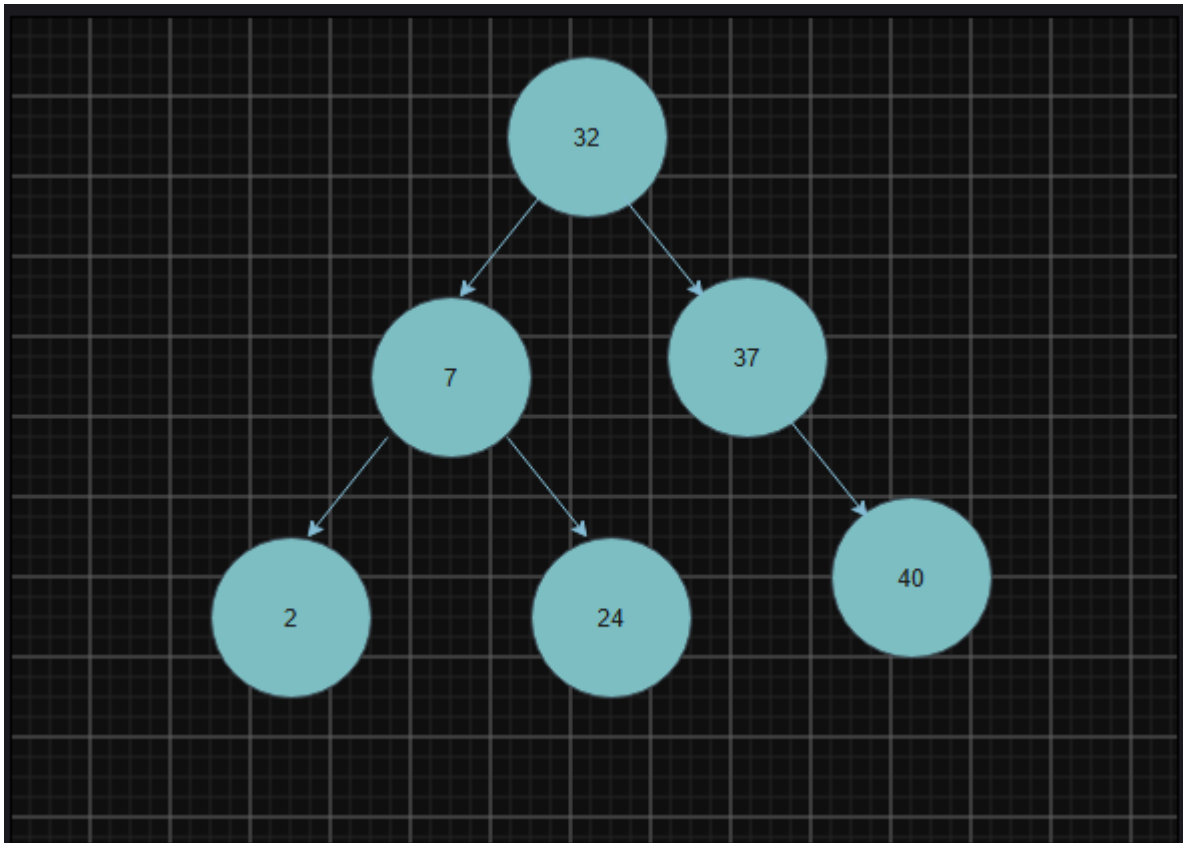
Diff 1. Aun no es necesaria la rotacion



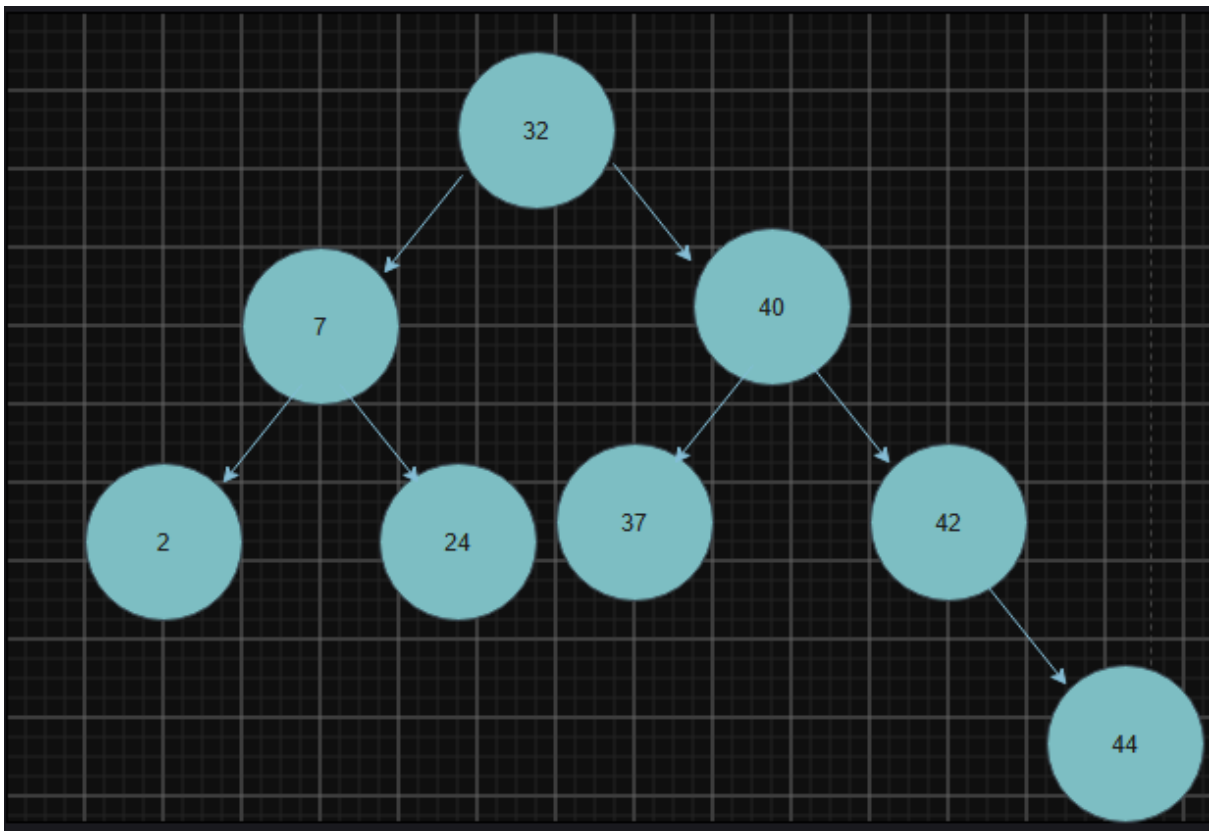
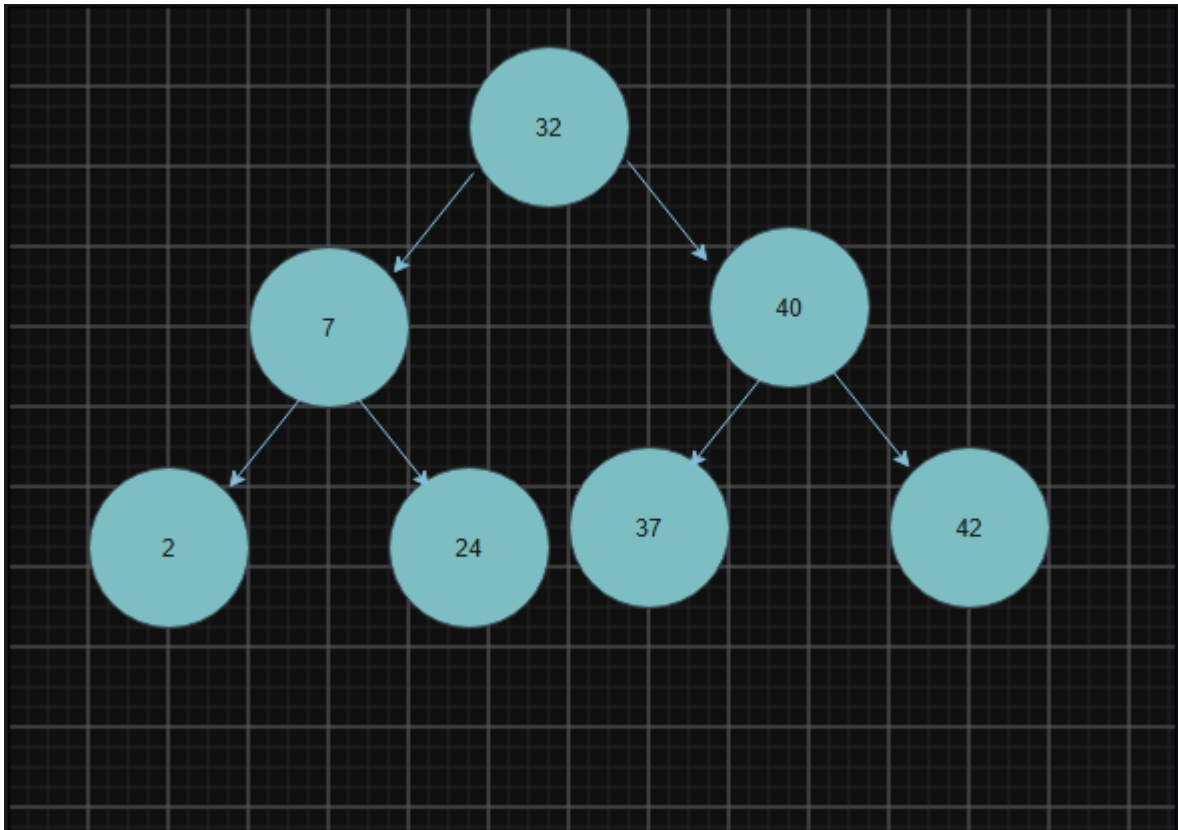
24 es el primer nodo desbalanceado.



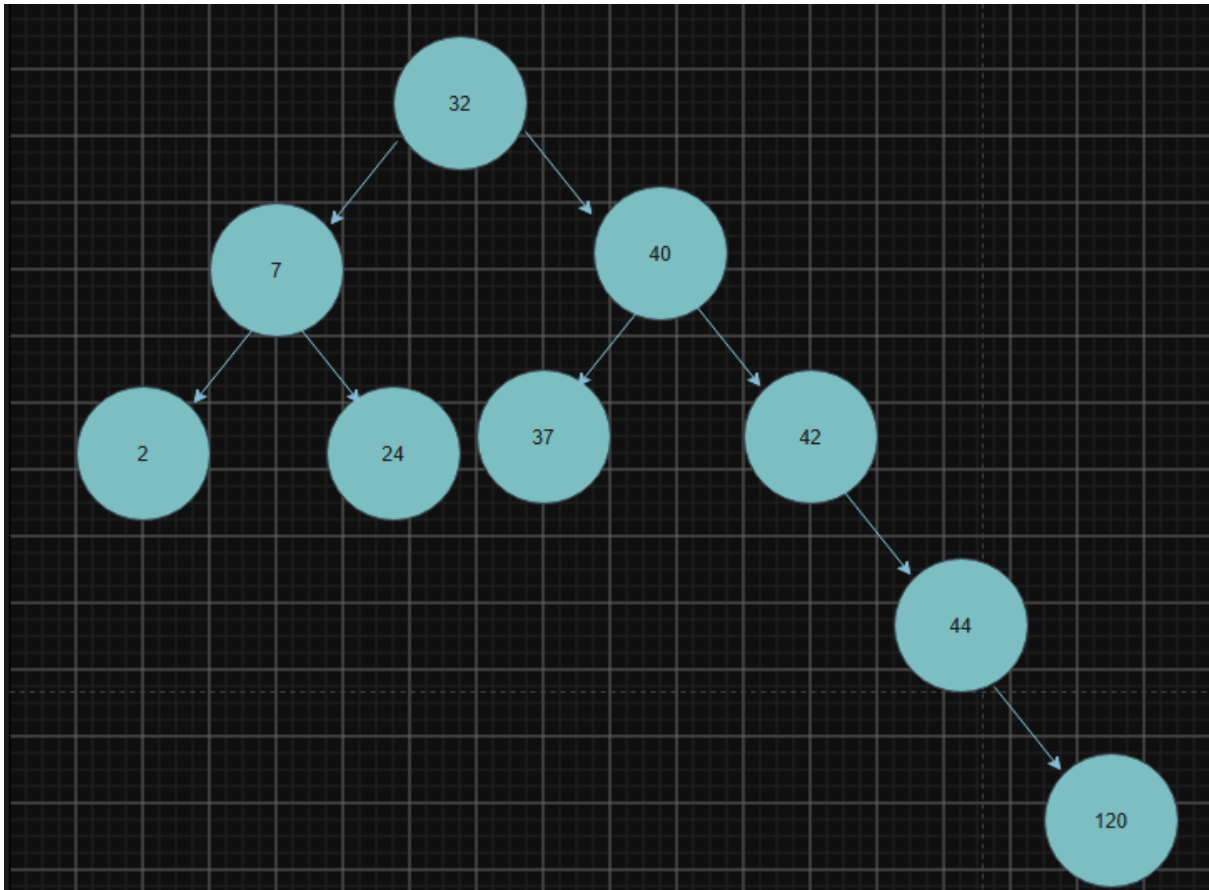
Ahora el desbalanceado es 7. Rotación Simple Derecha sobre 7



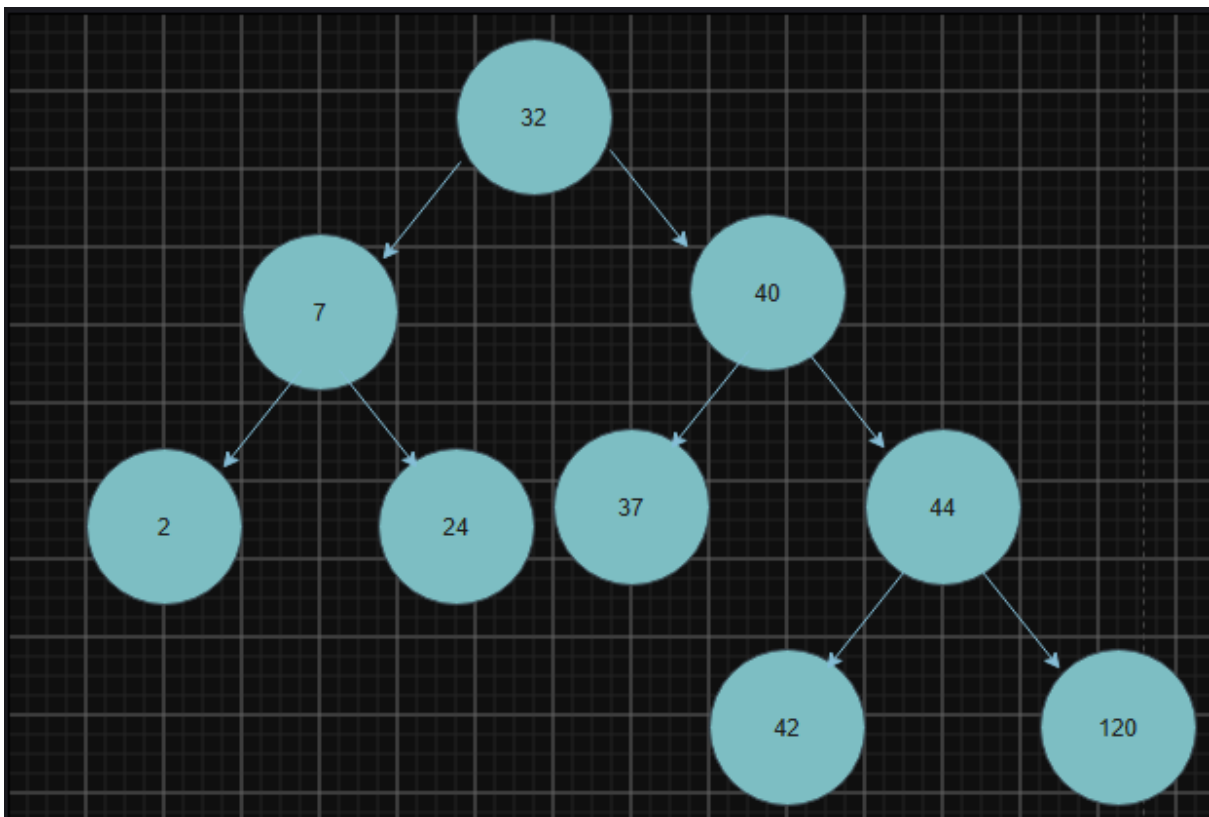
Diff 2. Esta desbalanceado. Rotación Simple Derecha



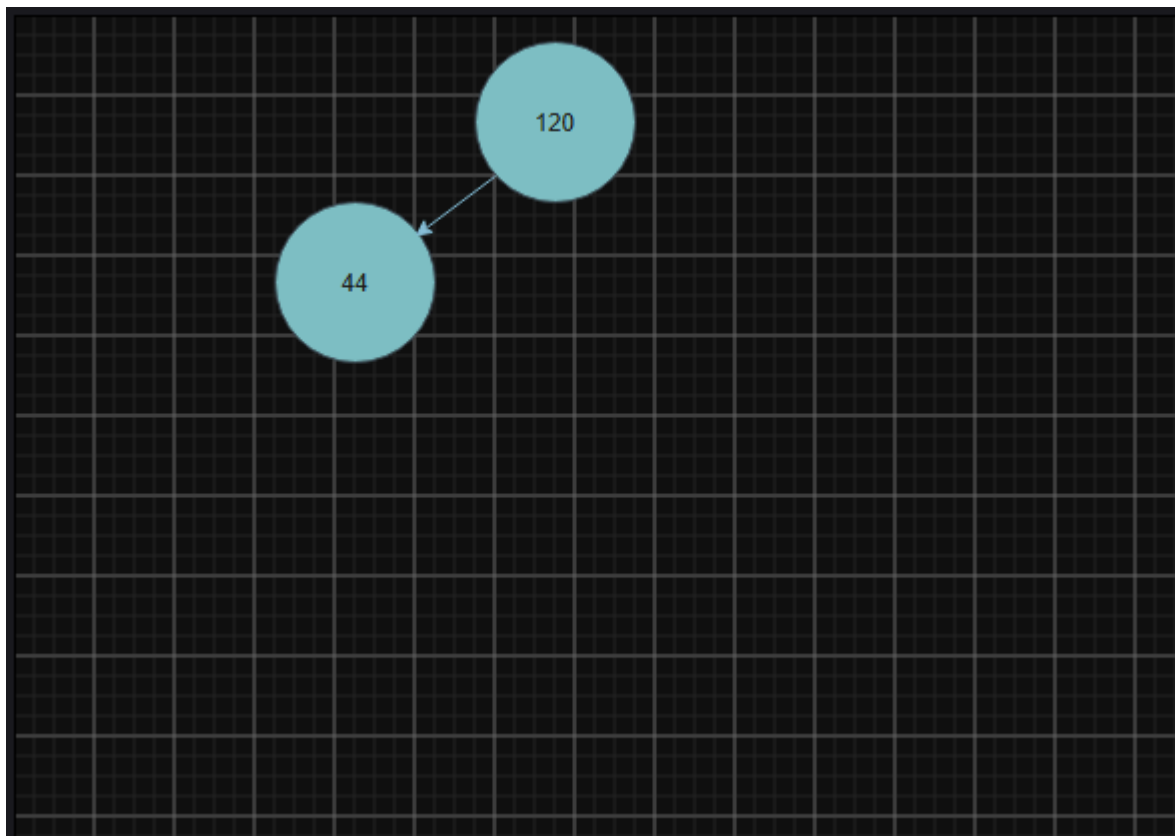
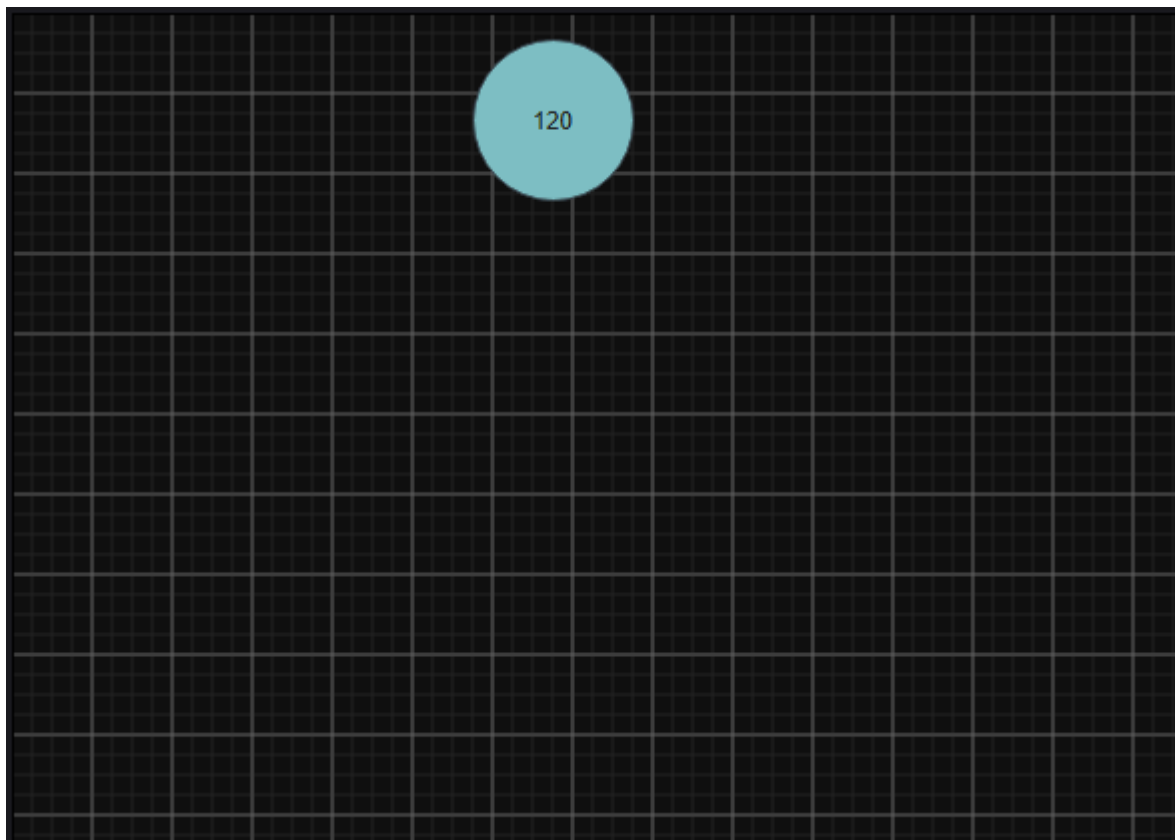
Diff 1.

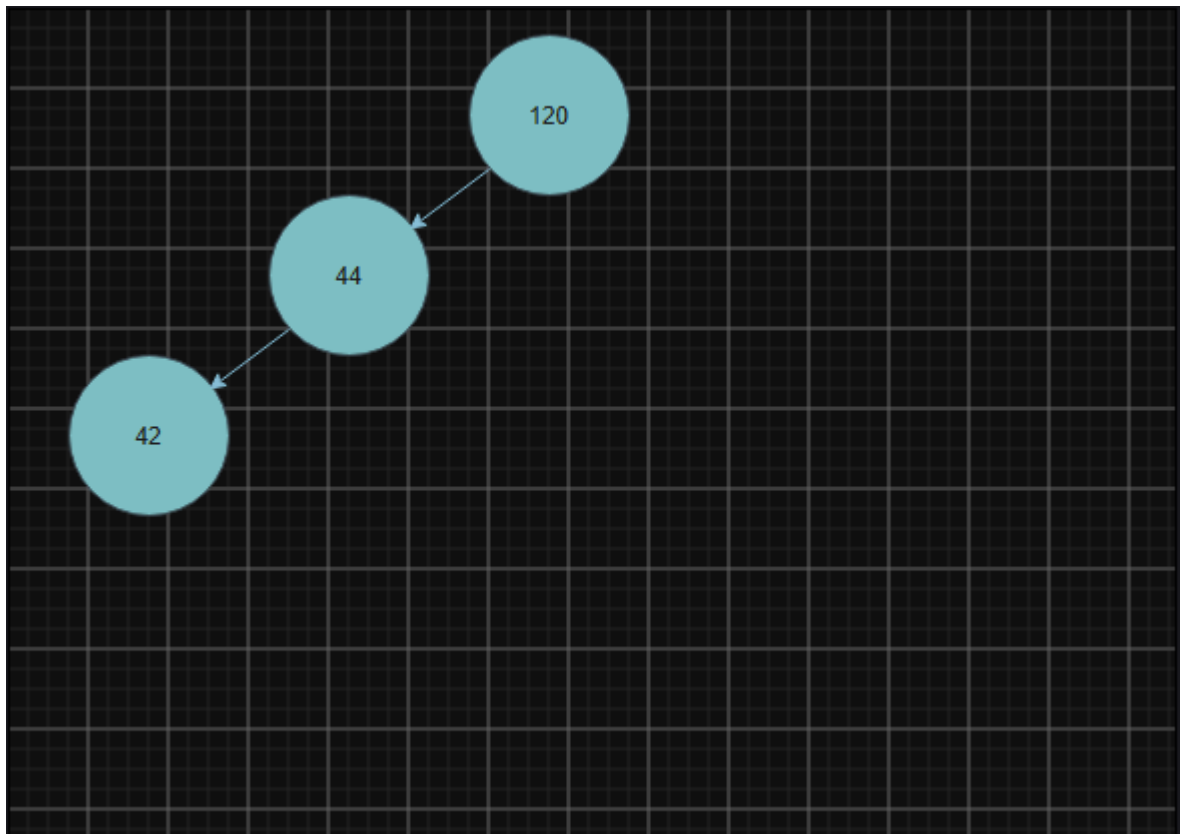


42 está desbalanceado

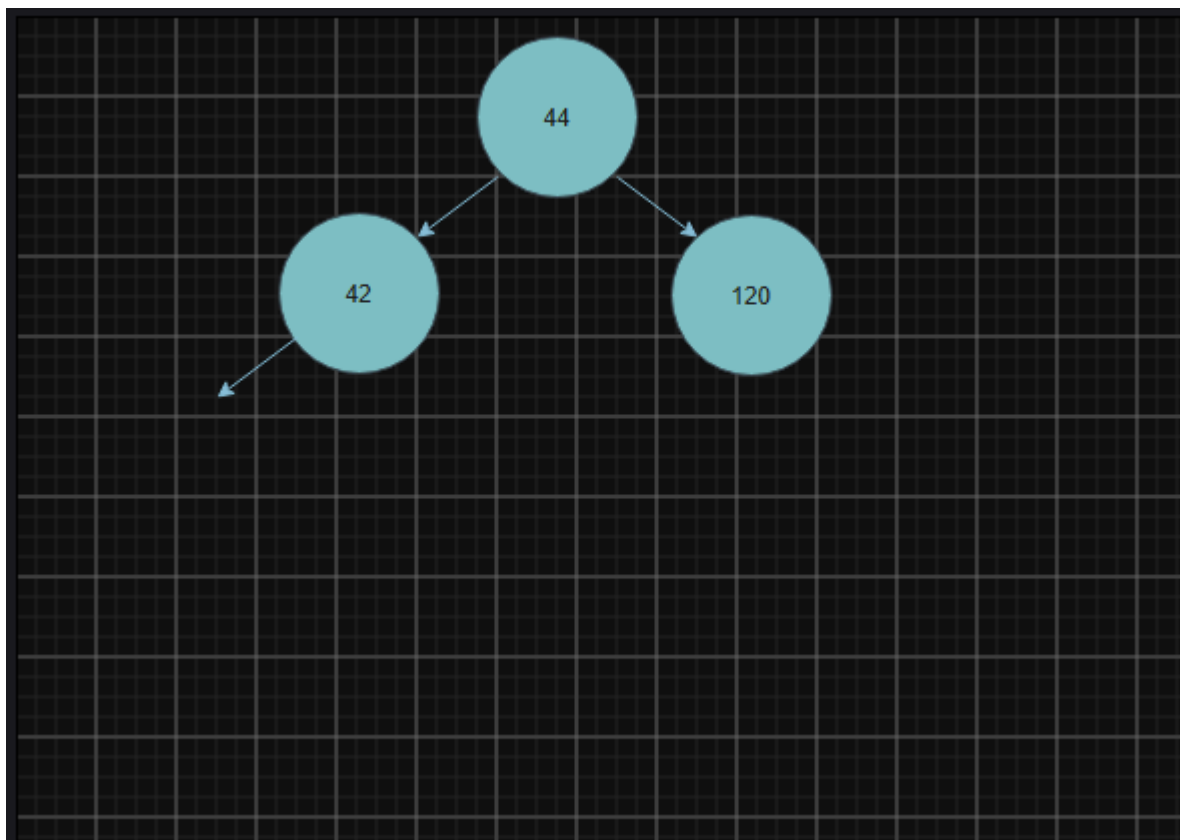


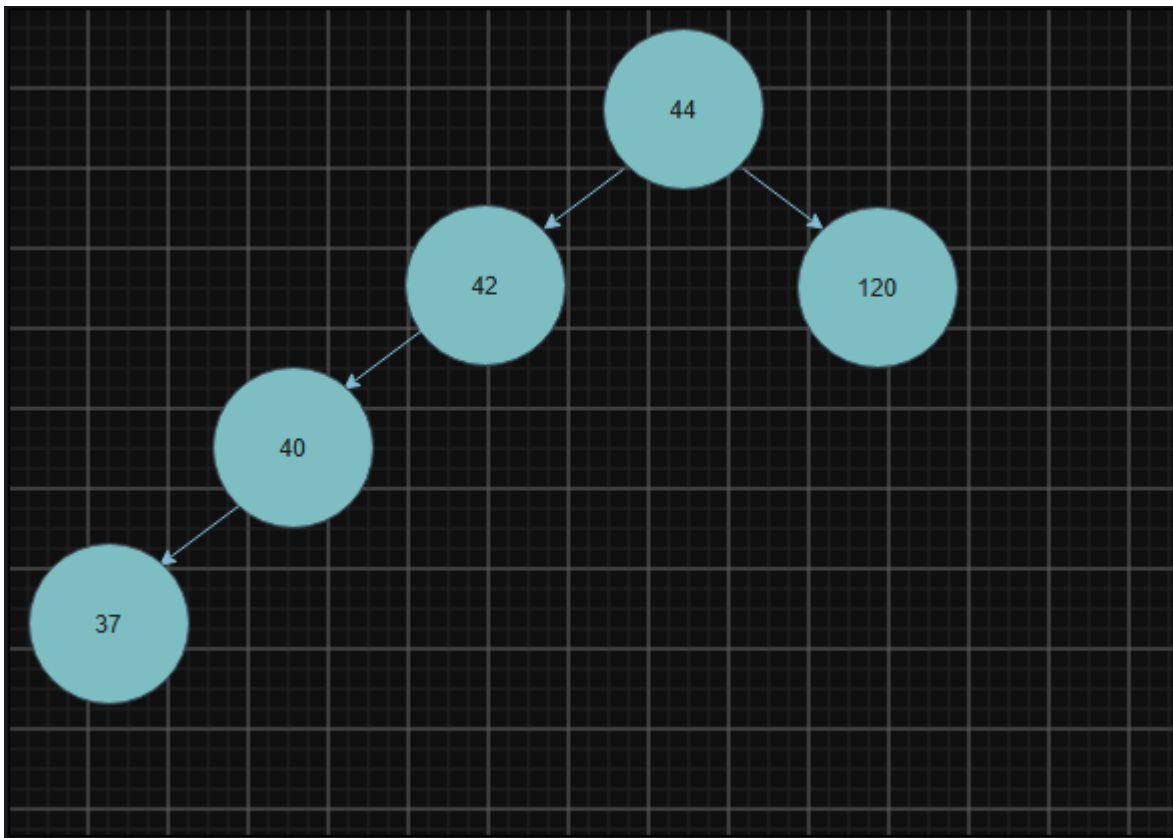
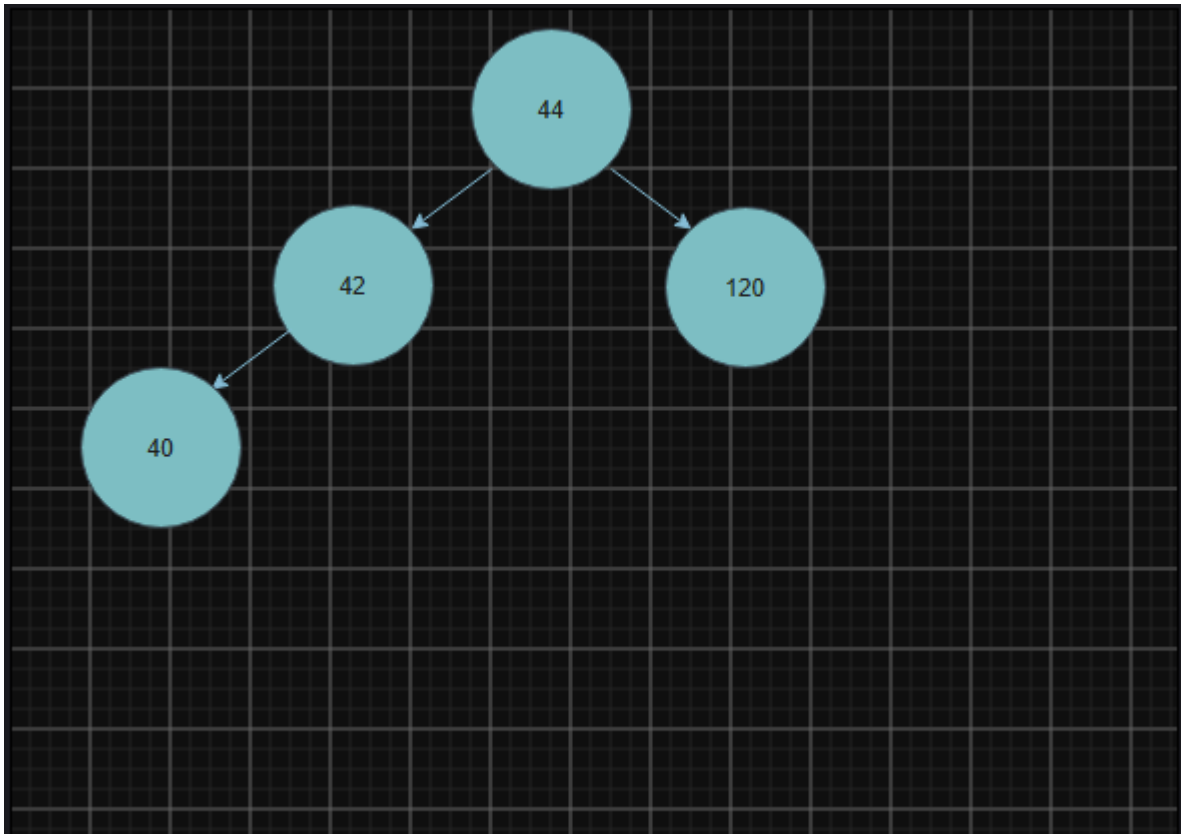
3.b.



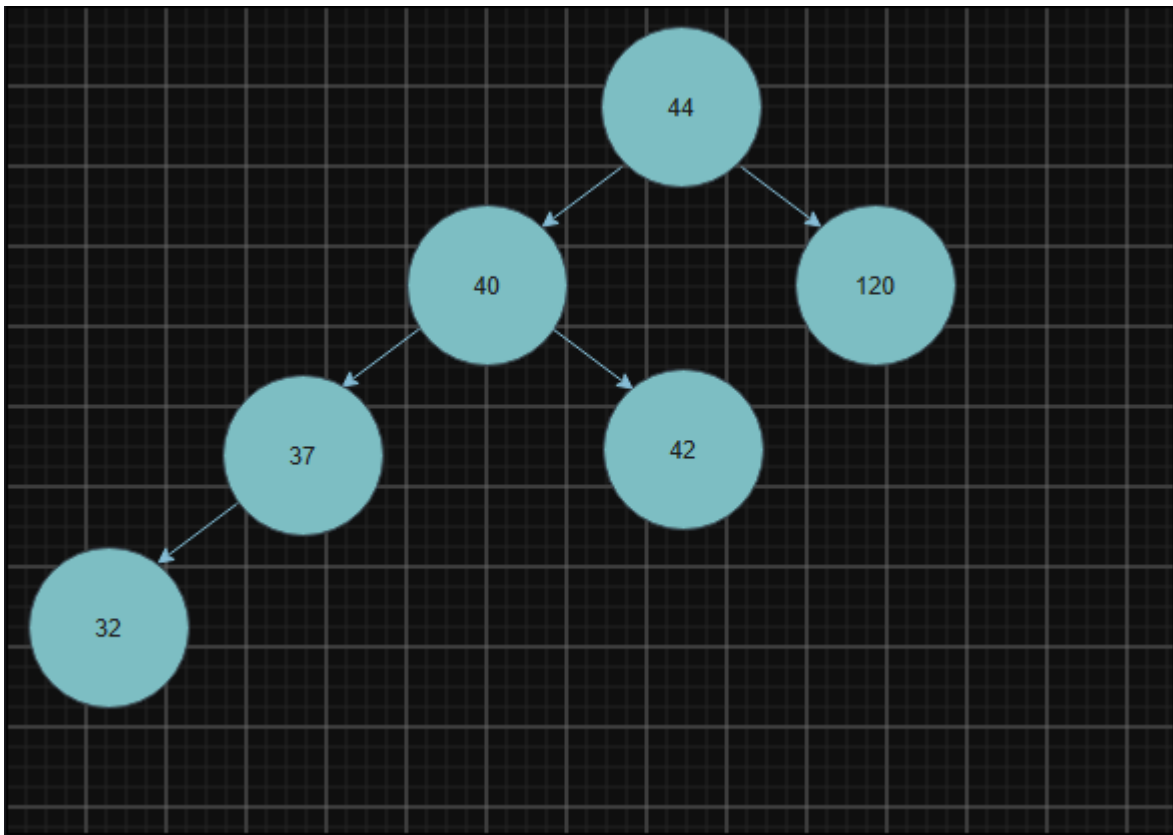
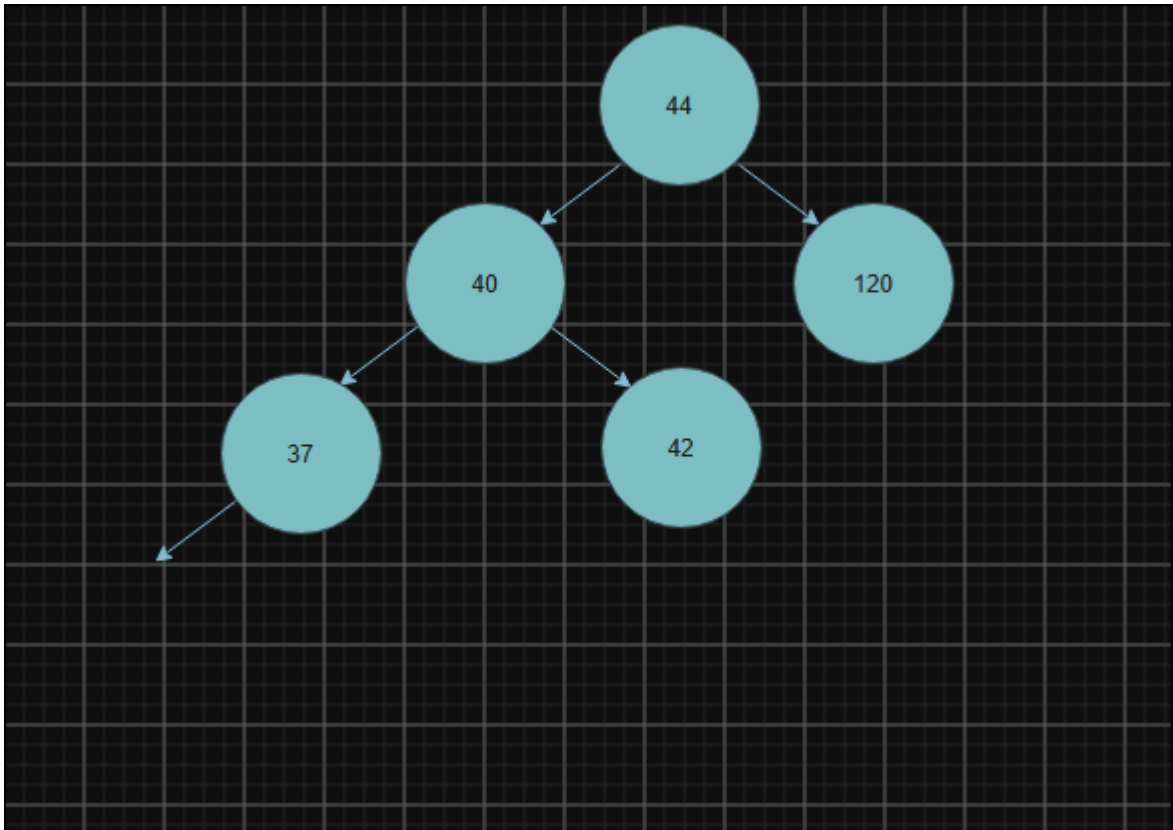


Rotación Simple Izquierda sobre 120

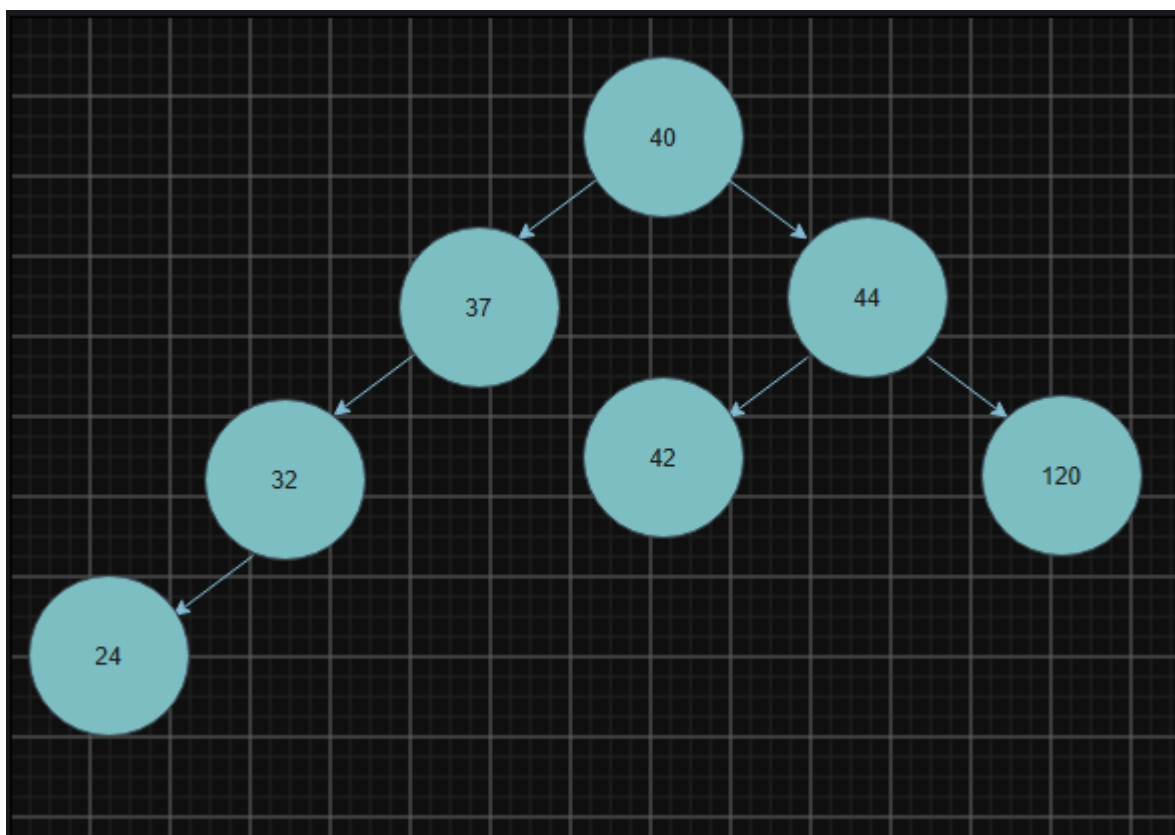
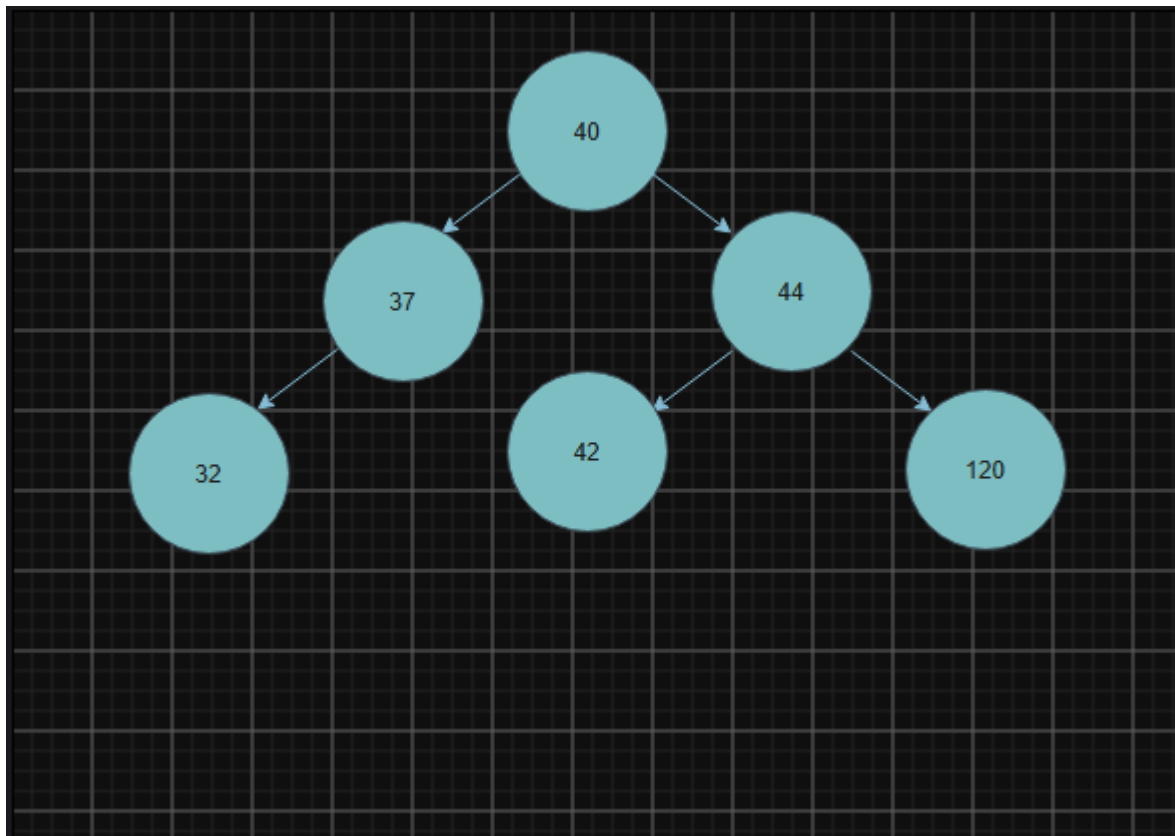




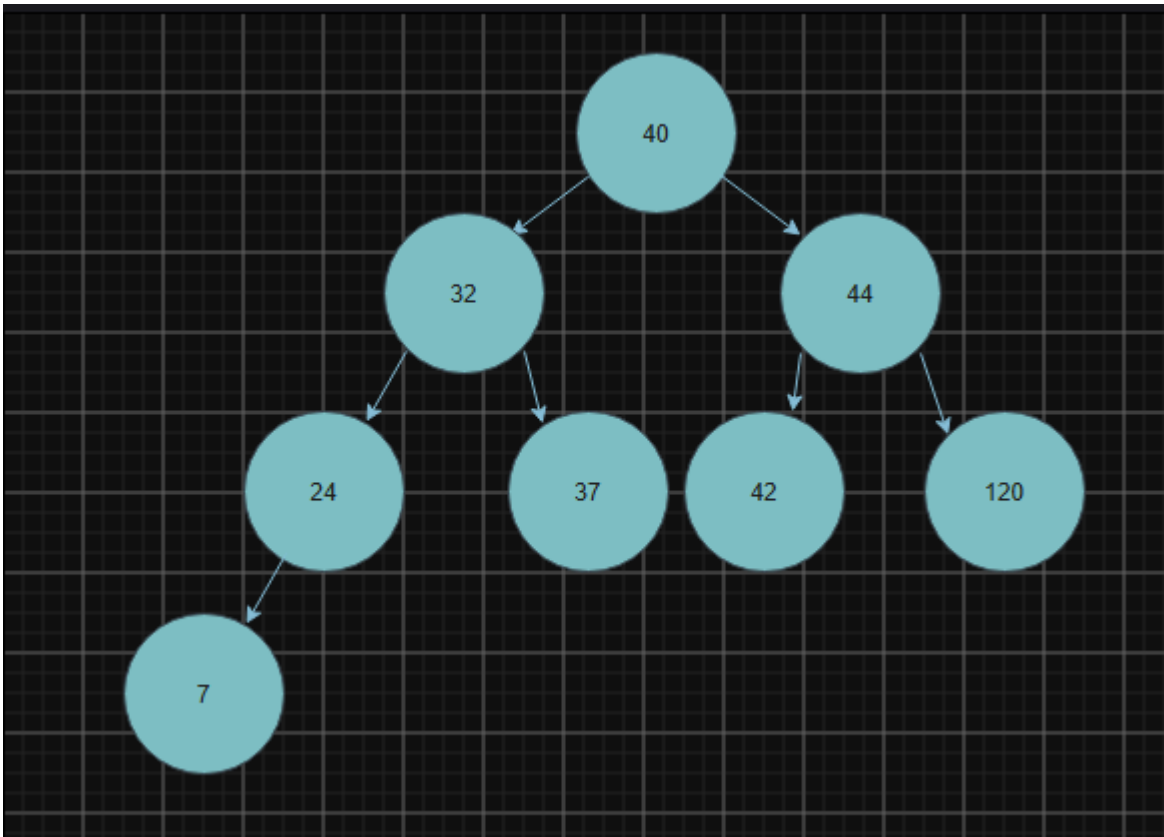
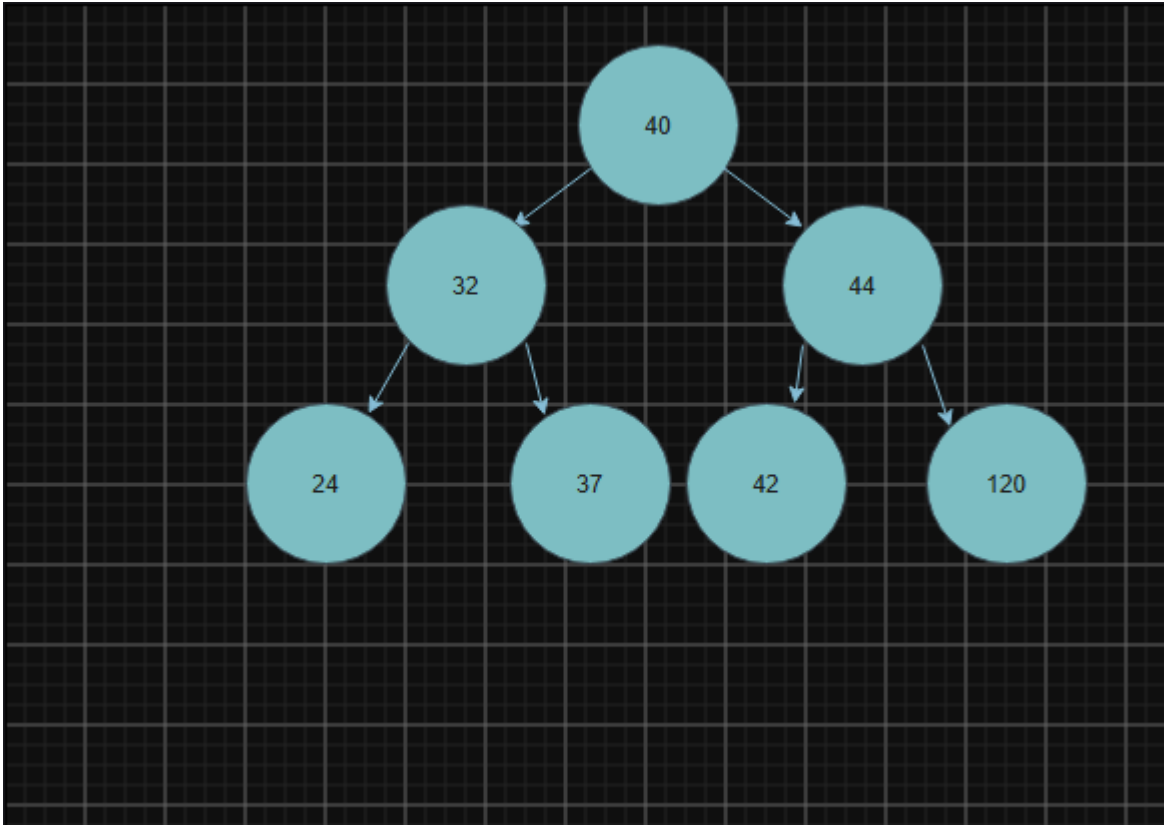
Rotación Simple Izquierda sobre 42.

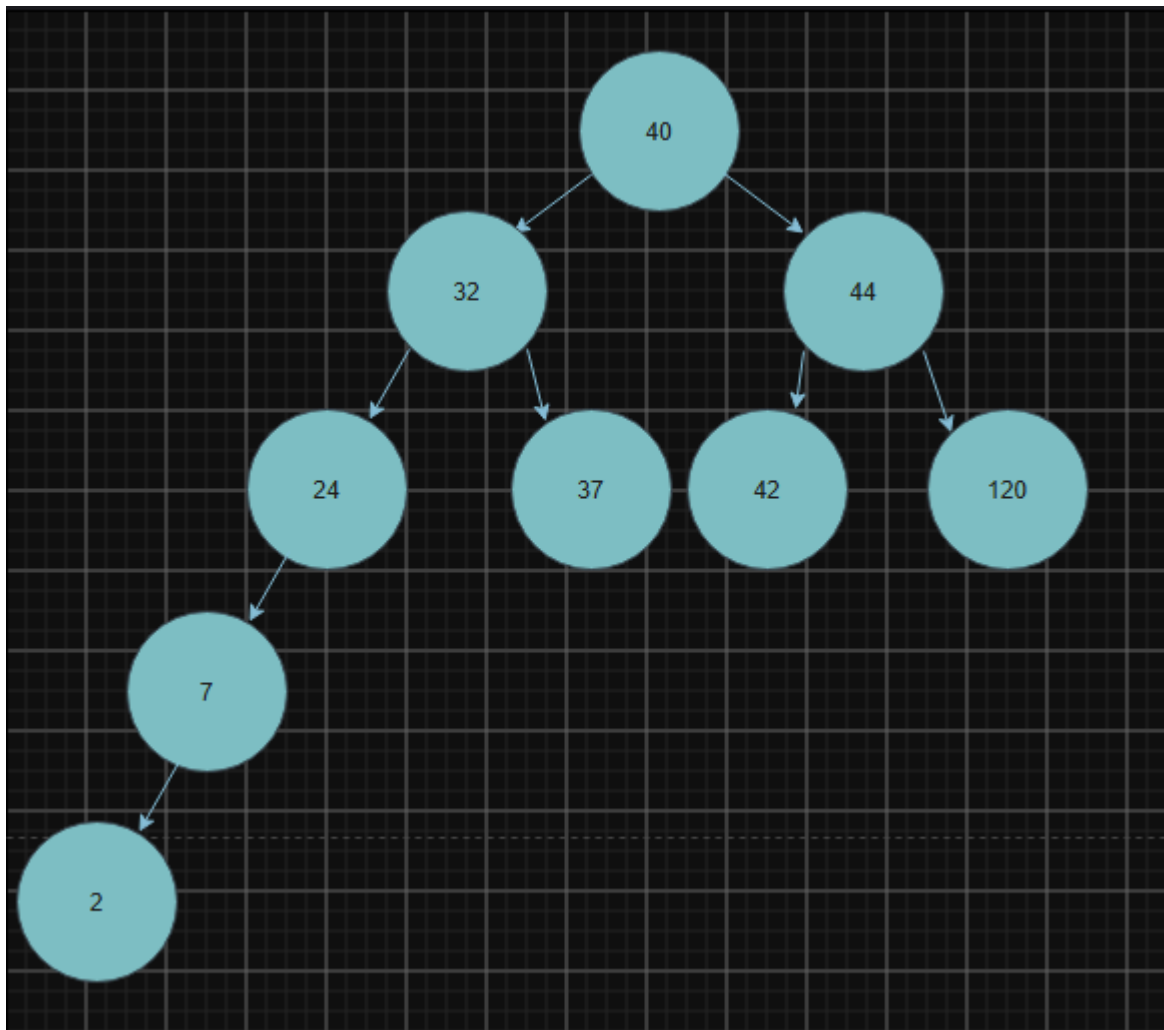


Rotación Simple Izquierda sobre 44.

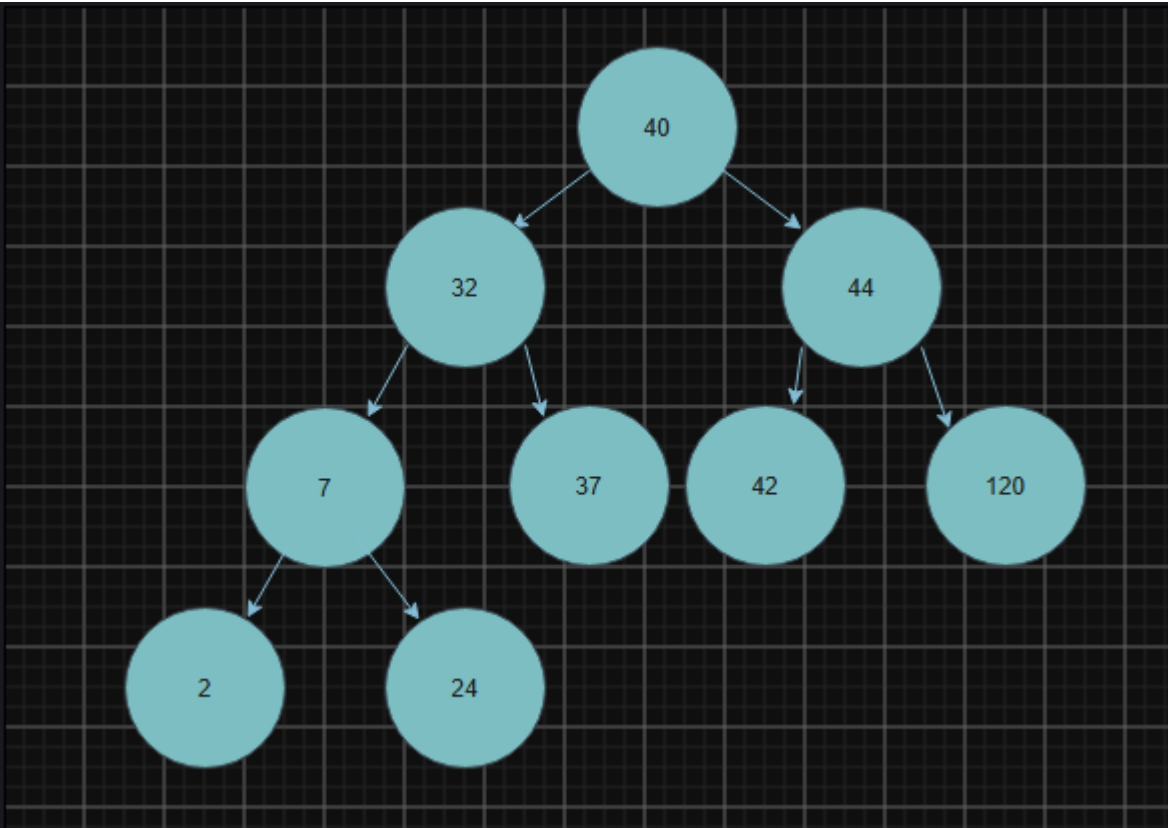


Rotación Simple Izquierda en 37

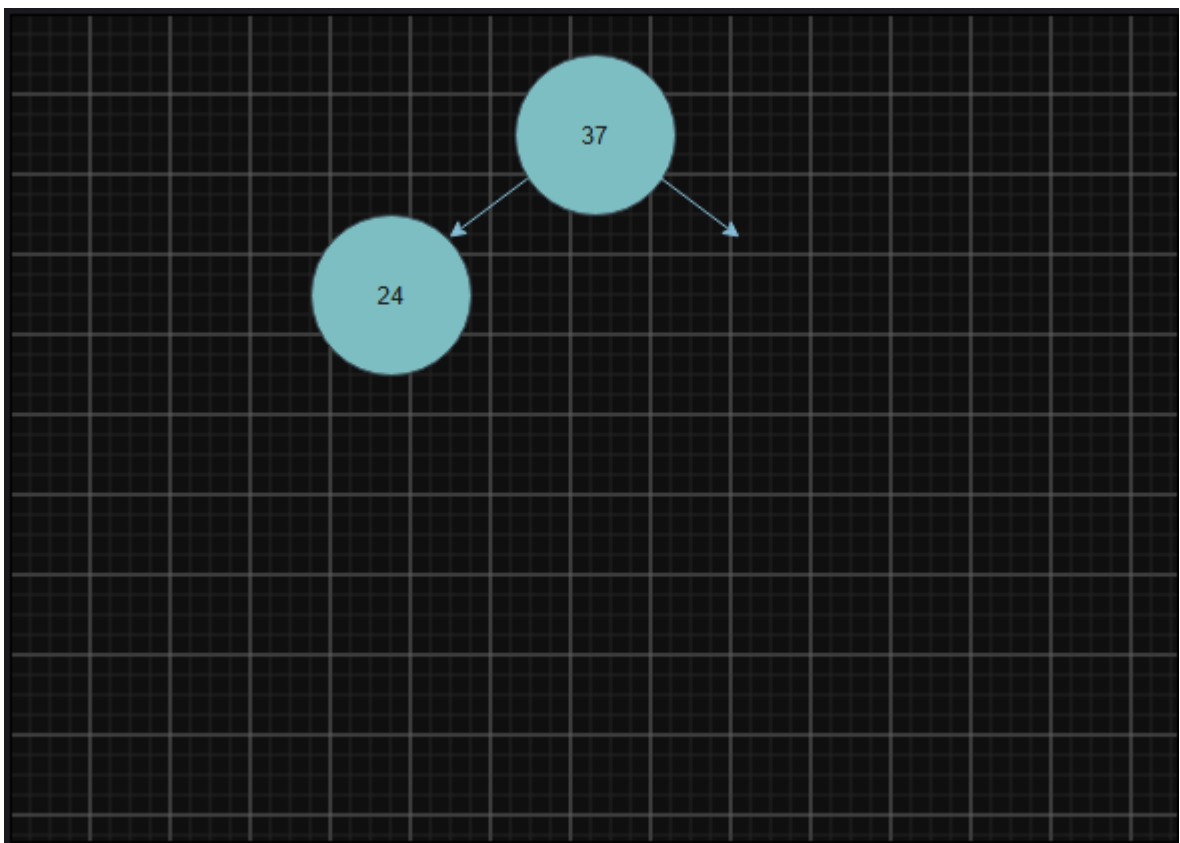
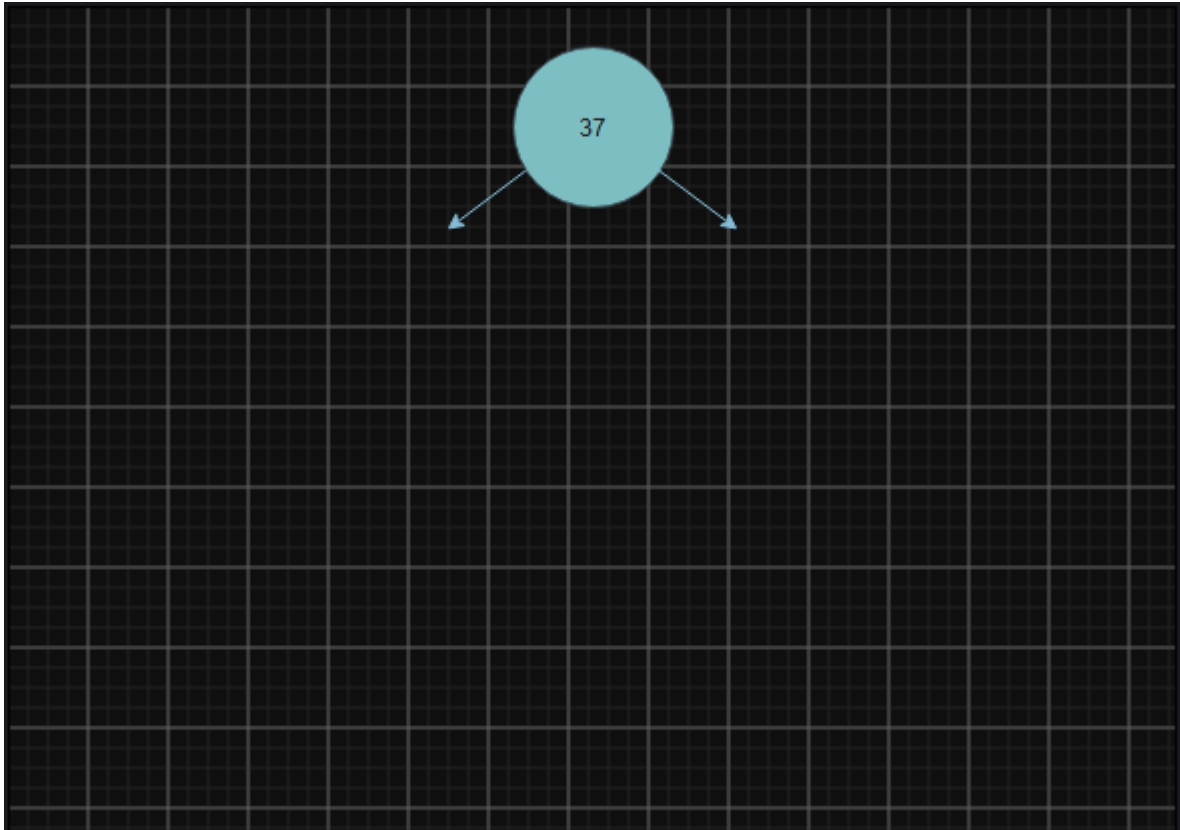


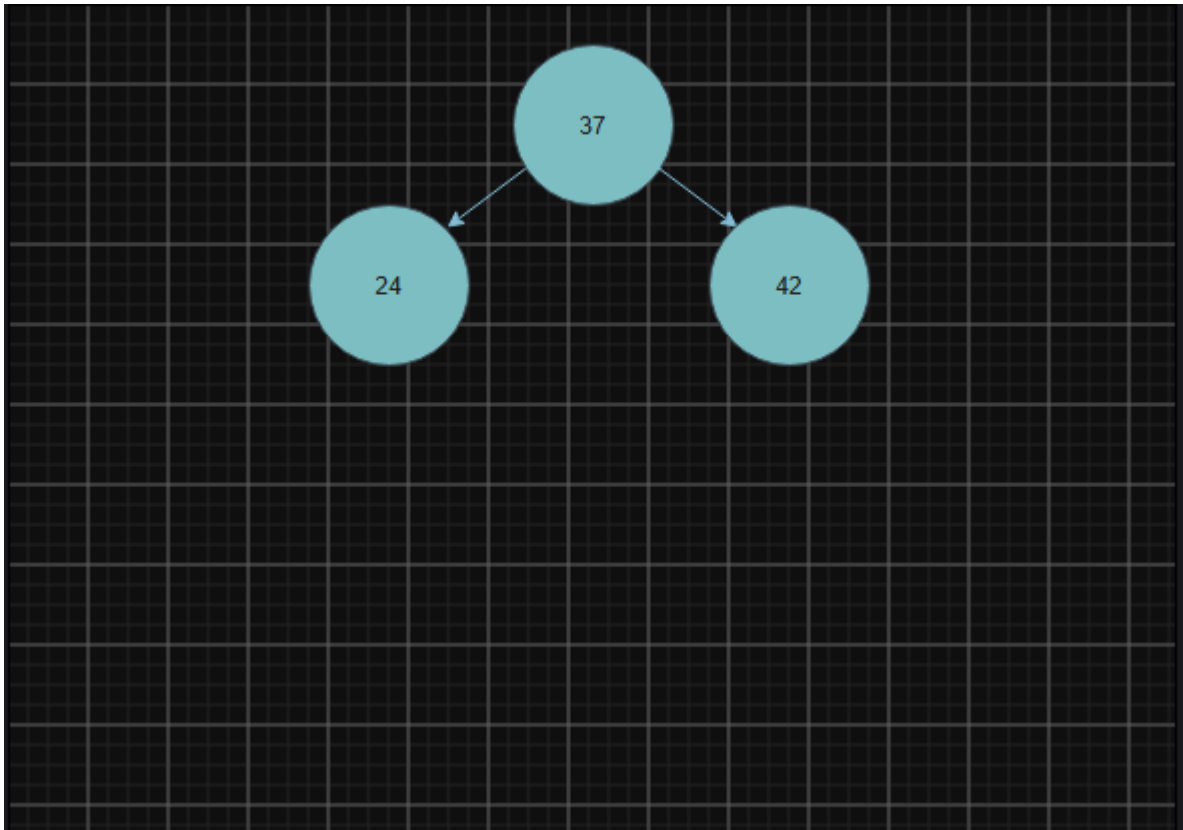


Rotación Simple Izquierda sobre 24.

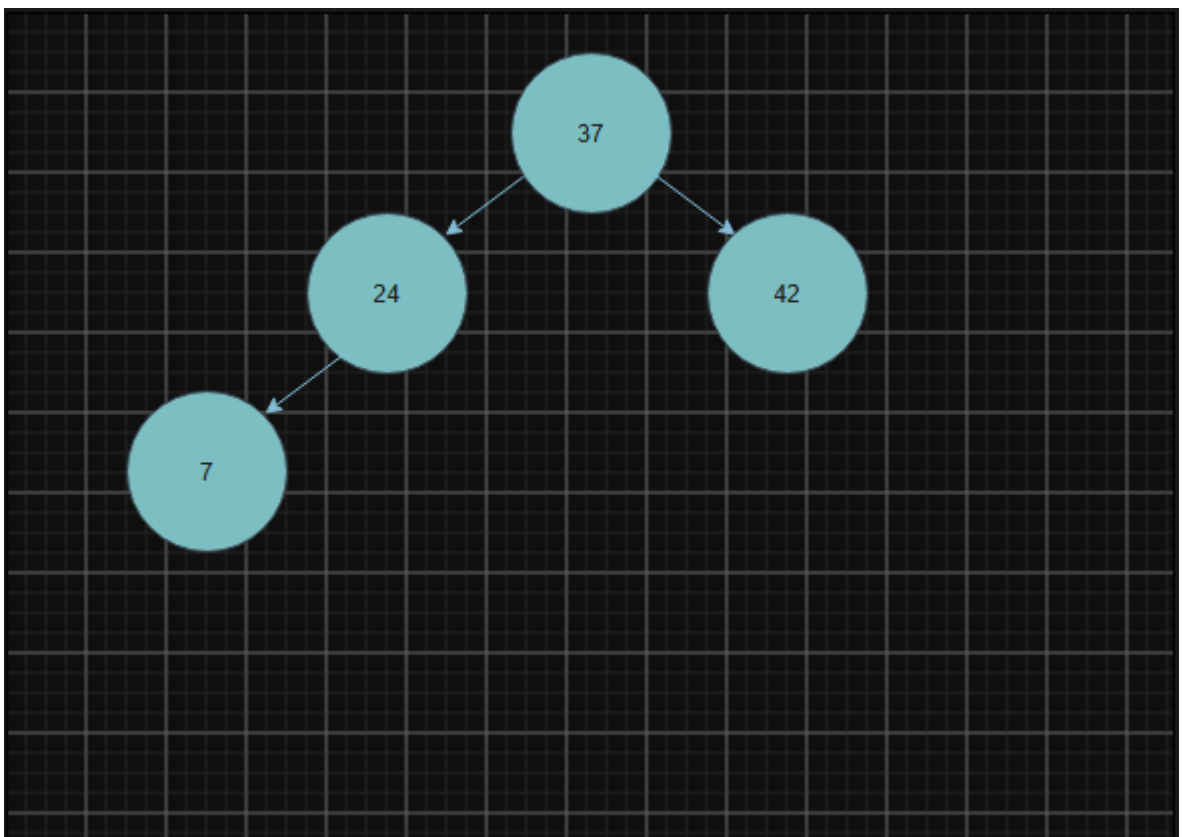


3.c.

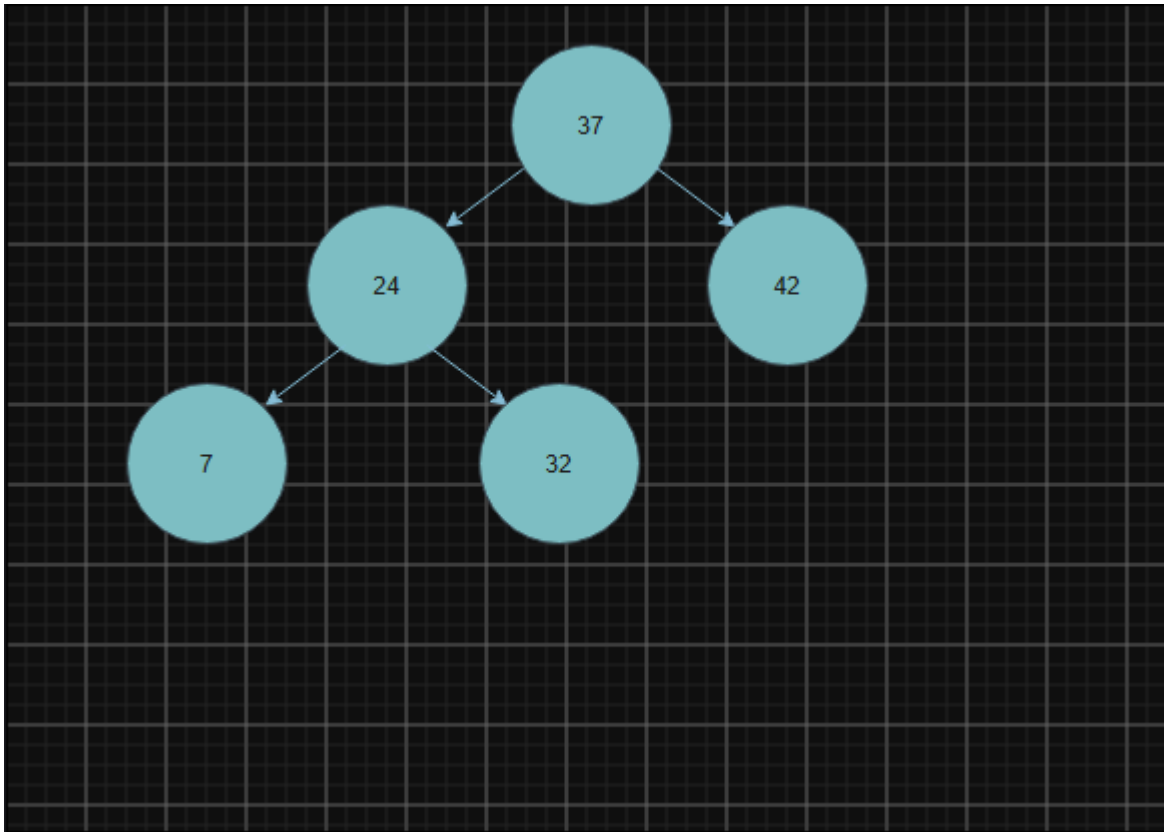




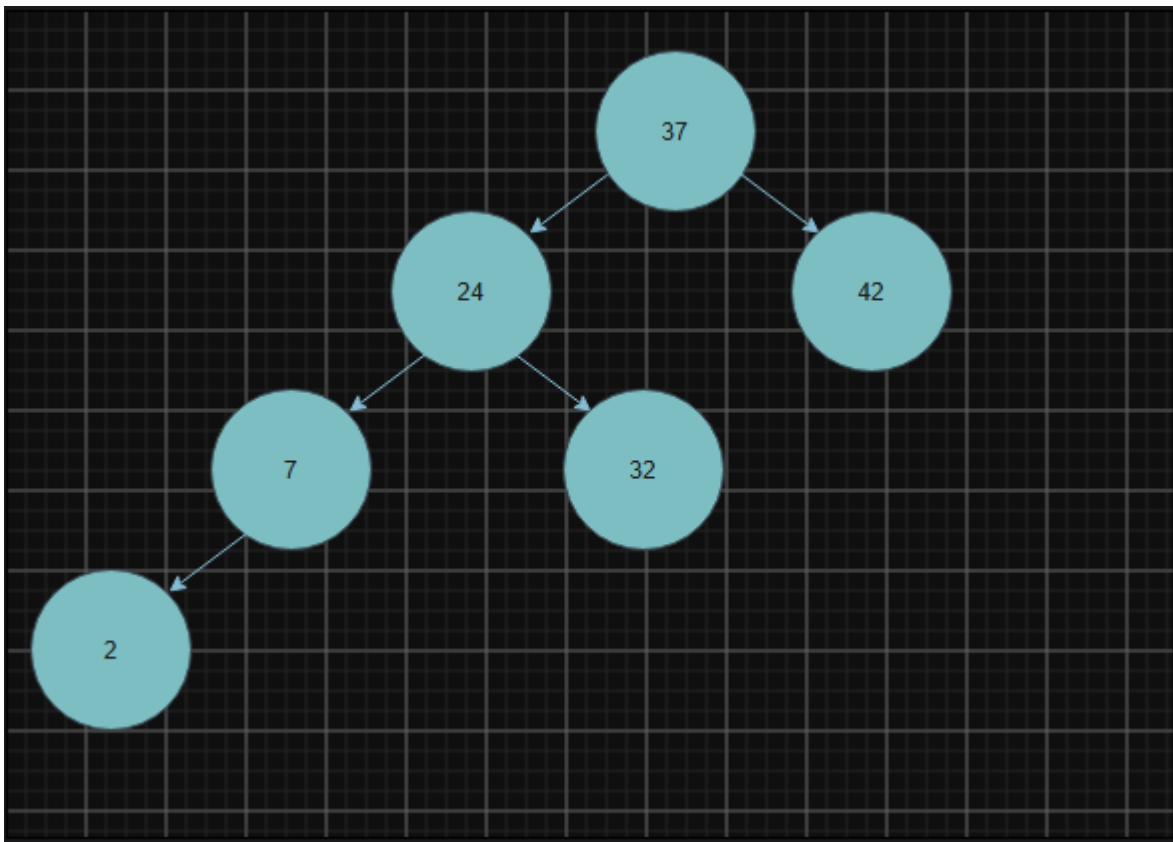
Sigue balanceado.



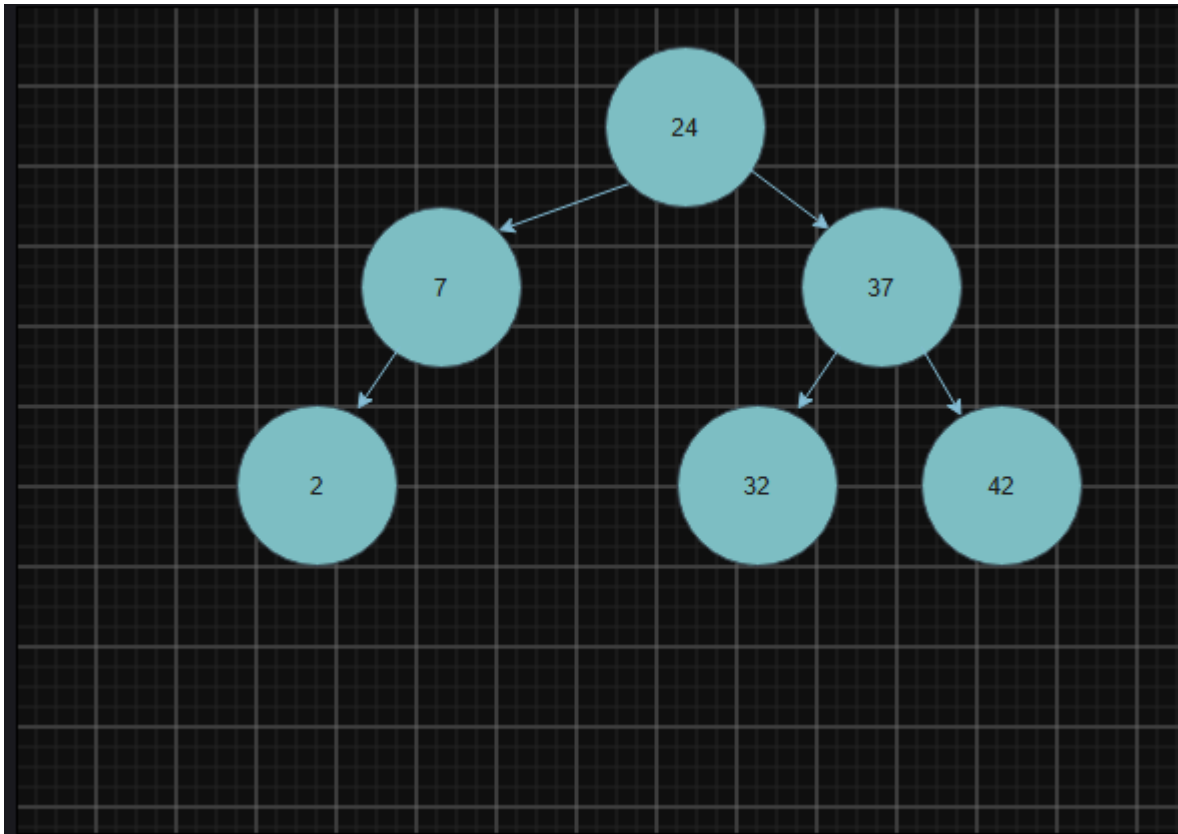
En 24 la diferencia es 1. En 37 también es 1. No se rota.



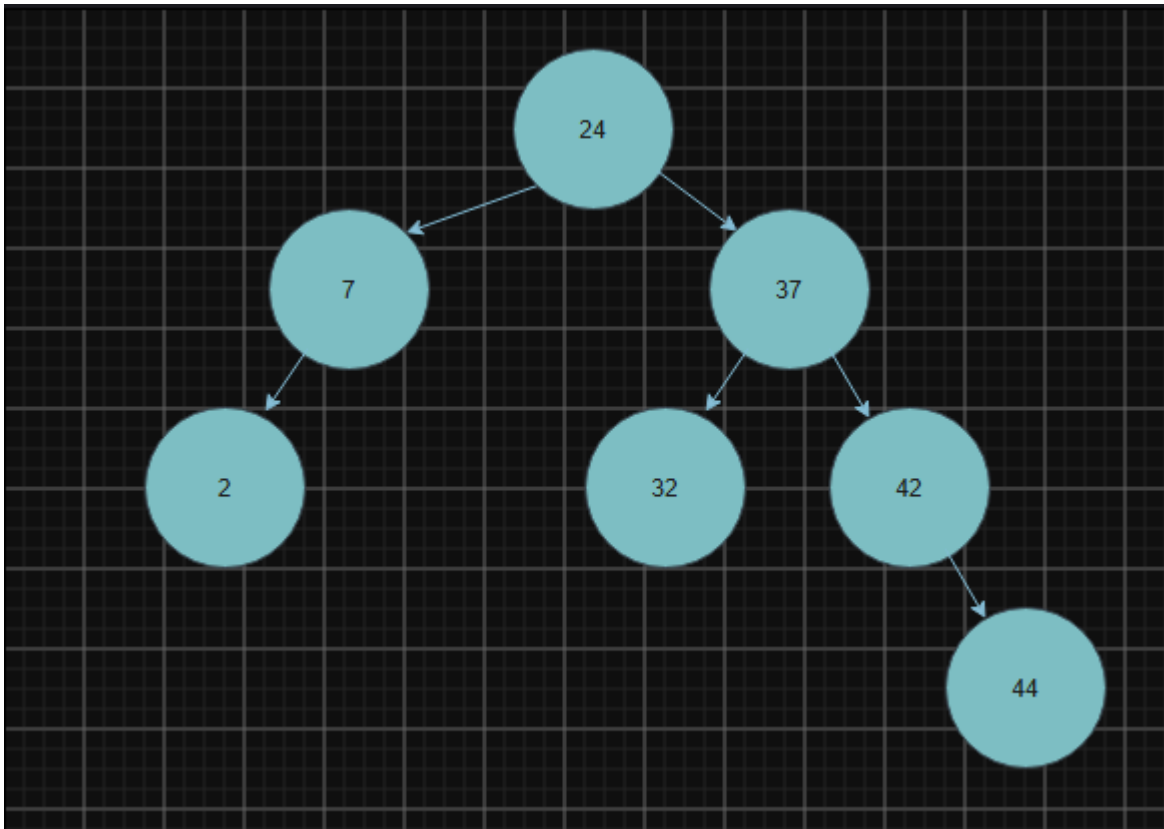
No se rota.



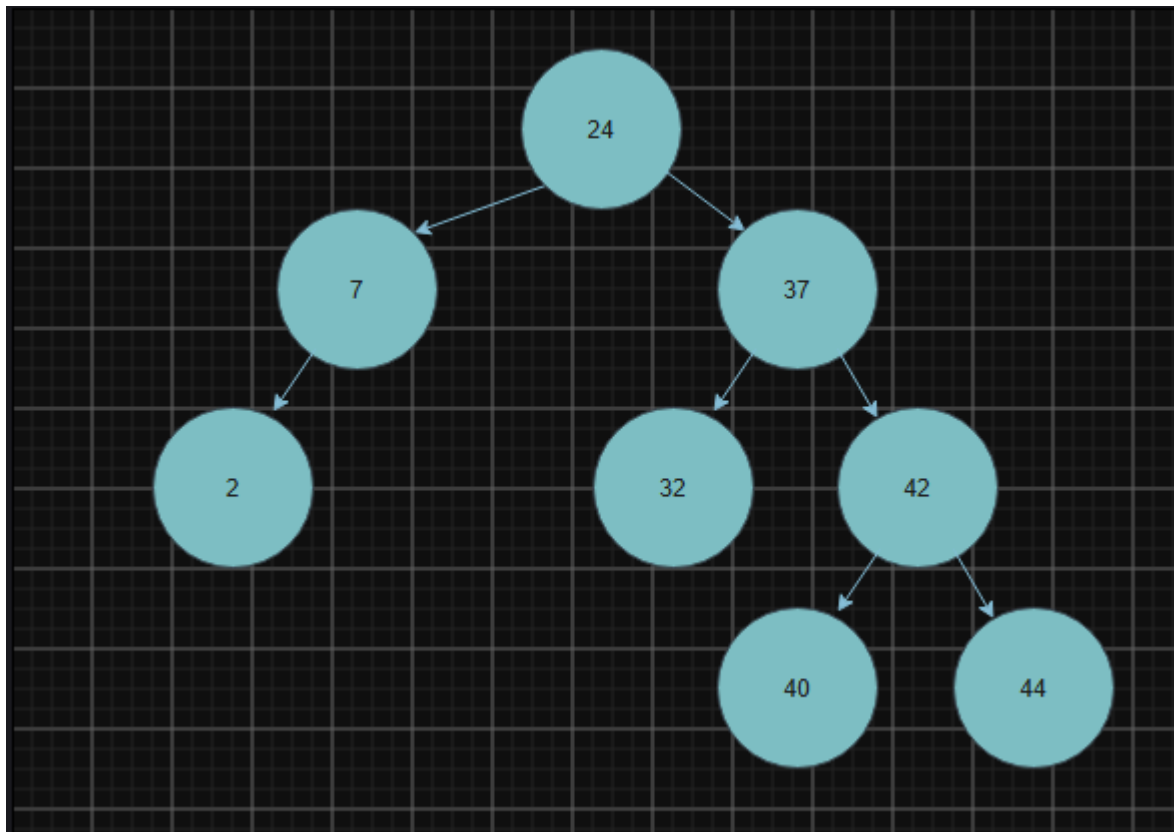
Su lado izquierdo tiene más altura que el derecho. 37 se desbalanceó. Rotación Simple Derecha.



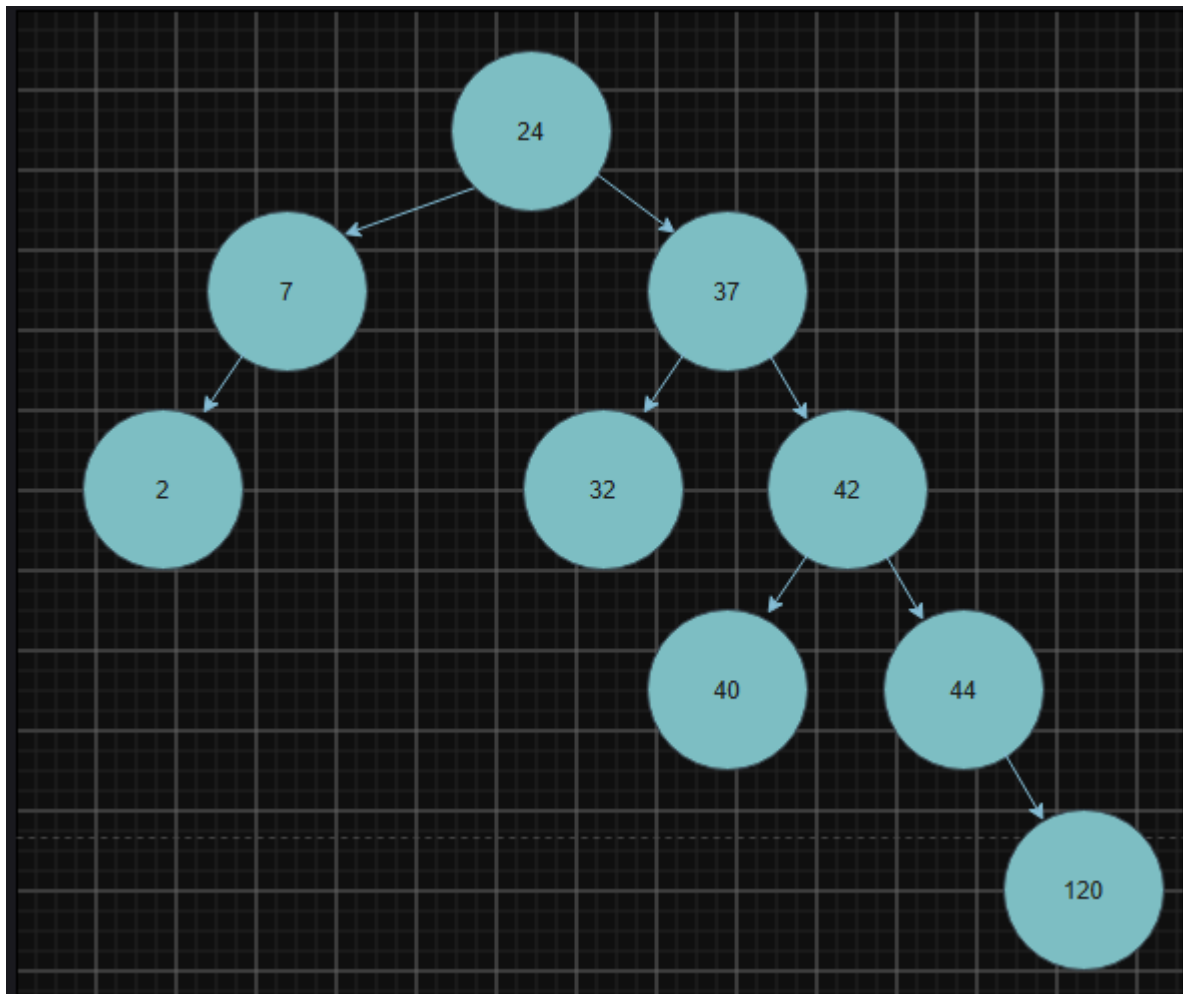
Para agregar 42 si agregar() encuentra un elemento igual, reemplaza el existente en lugar de crear otro nodo.



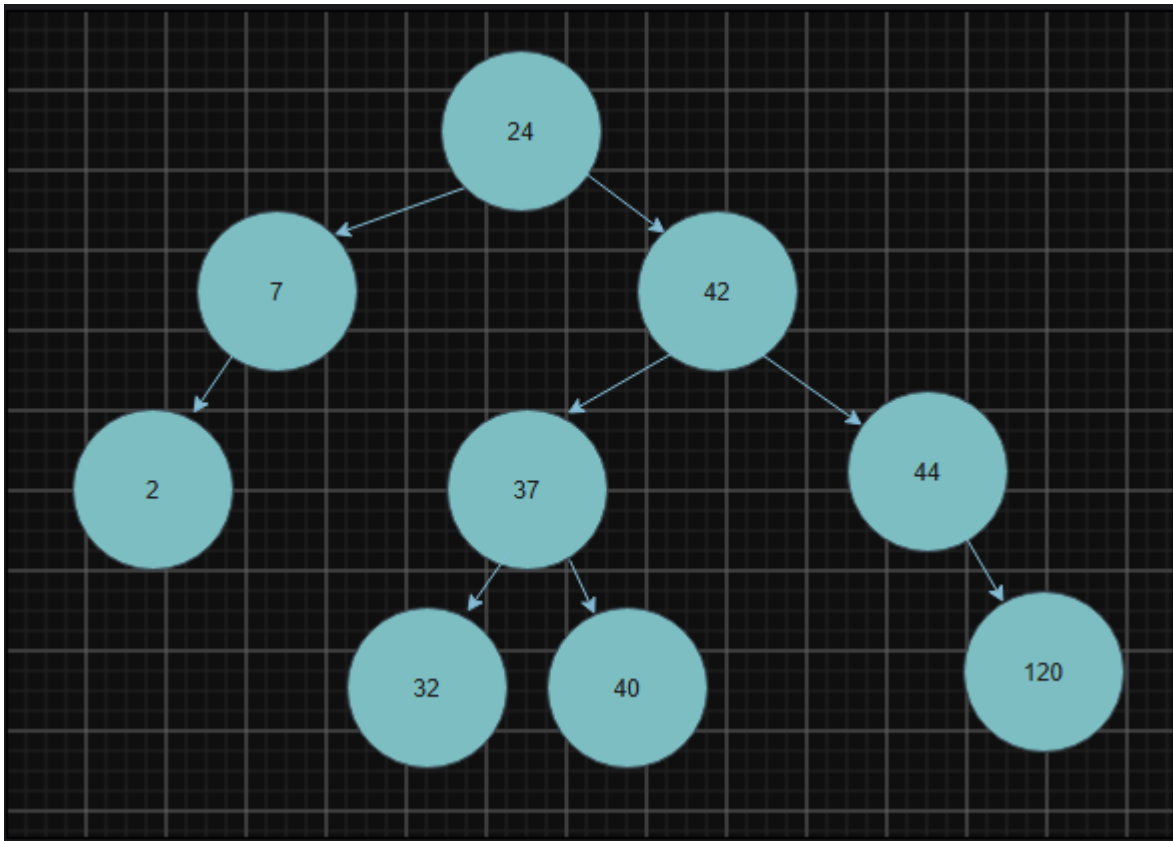
No requiere rotación



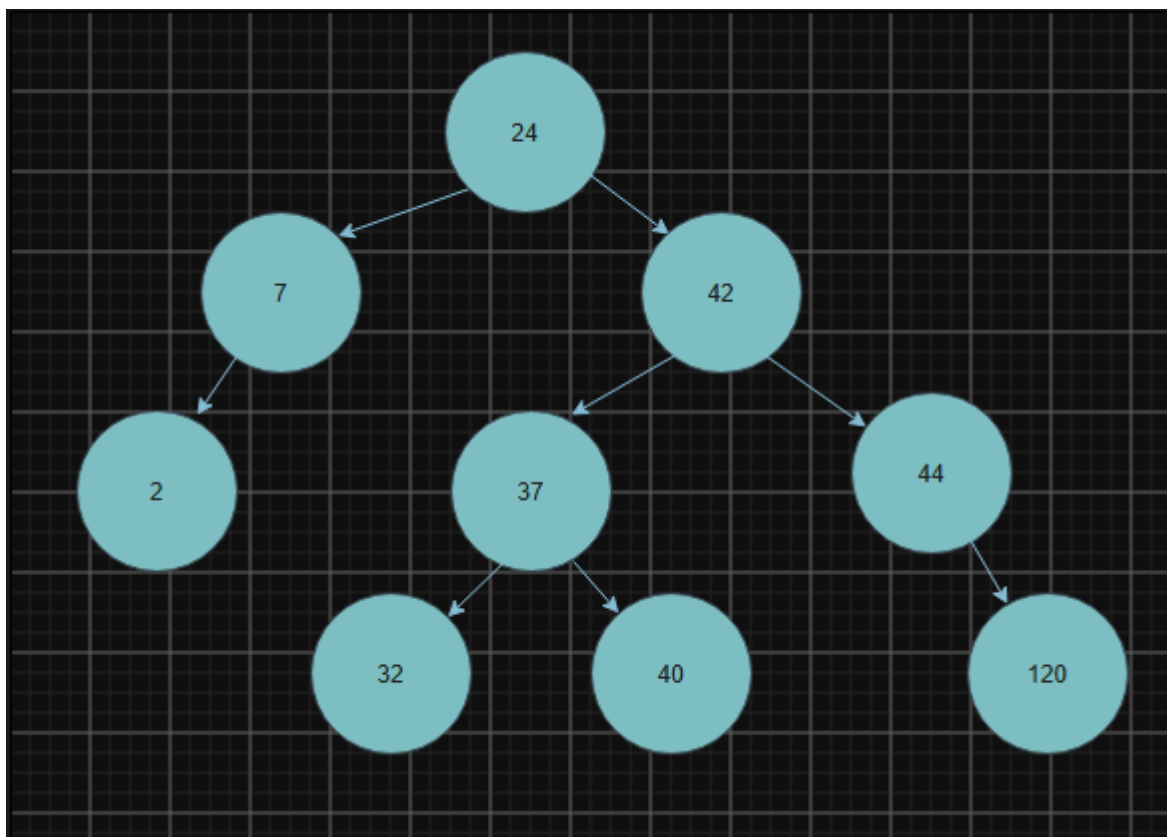
Pperfectamente balanceado



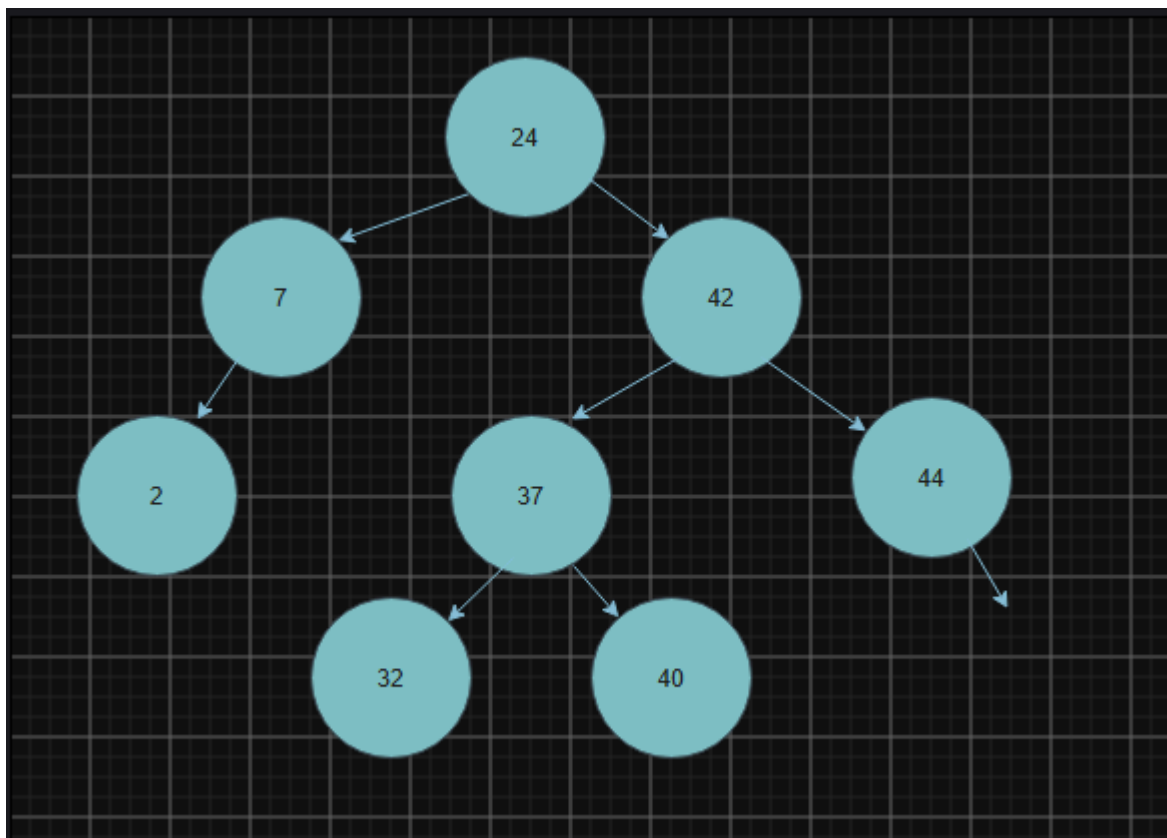
El lado derecho quedó dos niveles más alto que el izquierdo. Rotación Simple Izquierda desde 37



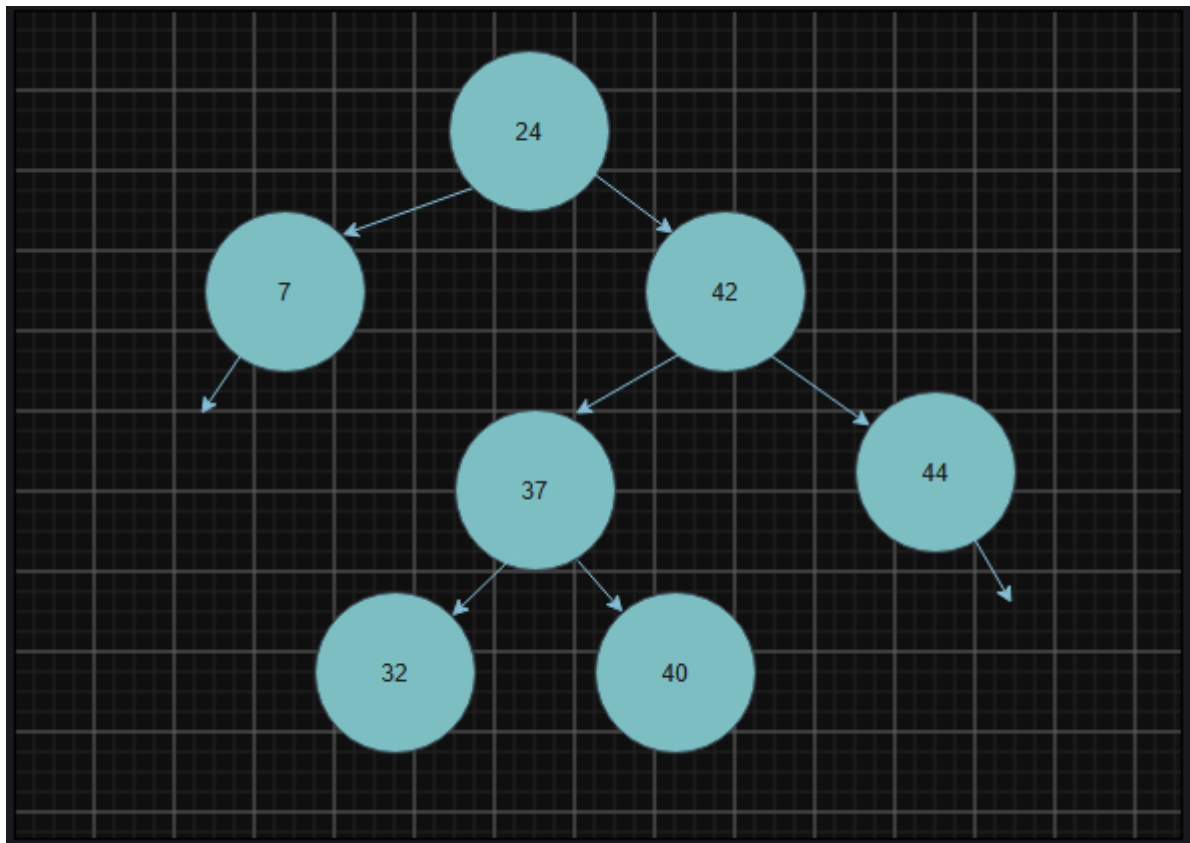
3.d



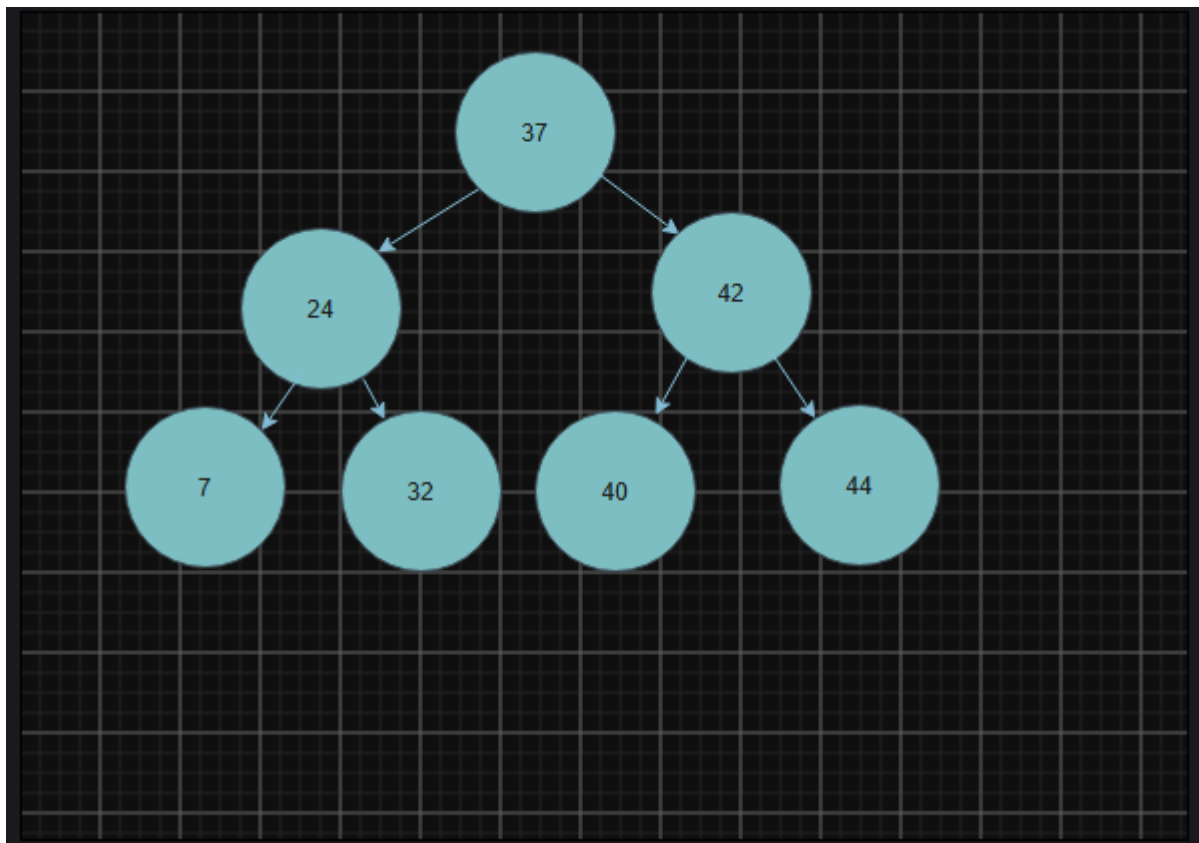
120 es una hoja.



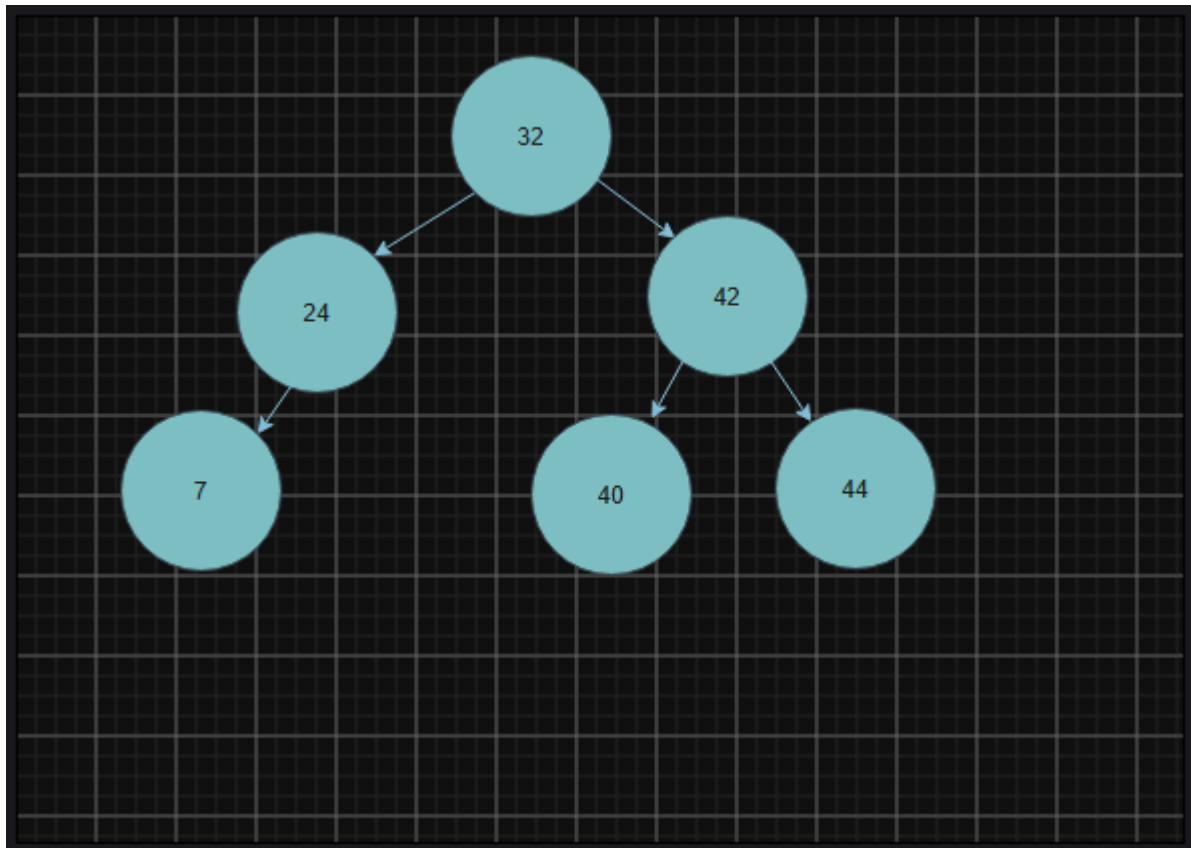
Sigue balanceado. 2 es hoja.



En 24 hay diferencia quedó desbalanceado. El árbol está cargado hacia la derecha de 24, requiere rotación doble.



37 no es hoja, es la raíz. Promuevo el mayor de los menores. El numero mas grande del subhijo izquierdo es 32, lo subo a la raíz.



No hace falta ninguna rotación.